

Summer Cooling Load Analysis and HVAC Design

The Grad House

University of Victoria

Prepared by: Luca Carinelli

Instructor: Lilah Montague

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1.0 Introduction

1.1 Project Background

The University of Victoria is undertaking a renovation of the Halpern Centre for Graduate Students (Building #160), a single-storey structure located on the main campus. The building previously housed the Graduate Students' Centre (GSC), providing offices, meeting rooms, and social space for graduate students. Under this project, the interior is redesigned to create an open-concept student social space featuring a restaurant and lounge, a bar, flexible meeting and multipurpose rooms, and a small office, open to the broader campus community.

This report presents the summer peak cooling load analysis and HVAC system design for the redesigned building [6]. The analysis supports the sizing of a new rooftop air handling unit (AHU), supply and return duct systems, and the selection of supply air diffusers for all conditioned spaces.

1.2 Building Description

Building #160 is a single-storey, wood-frame structure with an irregular floor plan and a peaked roof, located at the University of Victoria campus. The building footprint is approximately 32 m × 28 m. Structural walls, the existing kitchen, and washrooms are retained; all other interior partitions are removed and rebuilt as part of the renovation. The redesigned building contains 15 conditioned spaces totalling 466.97 m² of conditioned floor area. A new rooftop AHU is installed at the roof centre, above the mechanical room, as required by the project brief [6]. The existing

layout is shown in Drawing A1 (Existing Plan) and the proposed design in Drawing A4 (Interior Plan), both in Appendix A.

1.3 Project Objectives

The objectives of this project are:

- Perform a summer peak cooling load calculation for all conditioned spaces using the steady-state heat gain methodology from the ASHRAE 2013 Fundamentals Handbook [1].
- Design the supply air duct system and size all sections using the equal friction method.
- Select and schedule supply air diffusers for all rooms using the EH Price SPD catalogue [8].
- Design the return air duct system.
- Determine minimum outdoor air requirements per ASHRAE 62.1-2019 [5].
- Provide a basis for AHU cooling coil sizing and fan static pressure selection.

1.4 Scope of Work

This study covers the summer heat gain analysis only; winter heat loss is excluded.. Heat gain components addressed include: internal gains (occupancy, lighting, and equipment), opaque envelope conductive gains using the sol-air temperature method, window conductive and solar heat gains, and building infiltration [1]. The report covers the full HVAC design from load calculation through duct design and diffuser selection. The University's district energy hot water loop serves the AHU heating coil [6].

2.0 Building Geometry and Layout

2.1 Proposed Room Design Adjustments

The redesigned interior departs from the original room configuration in three specific areas, all intended to improve the functionality and spatial character of the new student space.

The partition wall between the vestibule (Room 101) and the restaurant and lounge (Room 110) was removed. This opens the building entry directly into the main lounge, creating a more fluid and welcoming transition from the entrance into the primary social space. The vestibule now functions as a threshold zone rather than a fully enclosed lobby. This change increases the effective floor area of Room 110 and is consistent with the open, social character of the proposed design.

The wall layout around Storage Room 108A was reconfigured so that the space is now accessed directly from Meeting Room 108 rather than from the main hallway. This gives the meeting room dedicated adjacent storage, useful for AV equipment and supplies. A secondary benefit of this adjustment is that it removed a partition from the north-facing wall of Room 110, increasing the usable perimeter wall length in the restaurant and lounge.

Storage Room 109 was extended slightly by absorbing a small portion of the adjacent corridor footprint. This consolidates storage in a single, slightly larger room and simplifies the circulation path through the central hallway zone.

These changes are modest in scale but improve the use of the space without altering structural walls or the retained kitchen and washroom areas. The demolition and construction extents are

shown in Drawing A2 (Demolition Plan) and the resulting layout is shown in Drawing A4 (Interior Plan) in Appendix A.

2.2 Space Classification (ASHRAE Space Types)

Each room is assigned an ASHRAE space type classification for the purpose of selecting lighting power densities from ASHRAE 90.1-2010 Table 7-16 and occupant heat gain values from ASHRAE 2013 Fundamentals Table 7-14 SI [1]. Classifications are listed in Tables 3 and 4. Storage rooms are classified as unoccupied for heat gain purposes, receiving only the minimum supply airflow required for conditioning.

2.3 HVAC Zoning

All conditioned spaces are served by a single rooftop constant-volume AHU with one supply duct system and one return duct system. No zone-level variable air volume (VAV) boxes or reheat coils are provided at this stage. This approach is appropriate for a building of this size, use type, and budget [6]. Room 107 is too small to warrant a dedicated supply run and is served by passive transfer air from Hallway 103 through a 200×100 mm transfer grille installed in the shared partition wall.

2.4 Floor Area Summary

Room dimensions were measured directly from AutoCAD drawings produced as part of this project. Table 1 summarizes all room floor areas, ceiling heights, and volumes [11].

Table 1: Room Areas, Ceiling Heights, and Volumes

Room No.	Space Type	Floor Area (m ²)	Ceiling Ht (m)	Volume (m ³)
101	Vestibule	5.73	2.75	15.8
102	Office	40.00	2.75	110.0
102A	Storage (adj. 102)	8.74	2.75	24.0
103	Hallway	19.52	2.75	53.7
104	Storage	1.23	2.75	3.4
105	Storage	0.54	2.75	1.5
106	Meeting Room	33.19	2.75	91.3
106A	Multipurpose Room	26.05	2.75	71.6
107	Storage	2.79	2.75	7.7
108	Meeting Room	15.48	2.75	42.6
108A	Storage (adj. 108)	9.48	2.75	26.1
109	Storage	14.29	2.75	39.3
110	Restaurant / Lounge	193.25	4.50	869.6
111	Bar	38.50	2.75	105.9
112	Multipurpose Room	58.18	2.75	160.0
TOTAL		466.97		1,622.5

3.0 Design Conditions and Assumptions

3.1 Outdoor Summer Design Conditions

Outdoor design conditions are taken from the ASHRAE 2021 Fundamentals Handbook, Table 1, for Victoria International Airport station [2]. The 1% annual exceedance dry-bulb temperature is used, which represents conditions exceeding no more than approximately 88 hours per year [2]. This is the ASHRAE standard basis for cooling system peak load sizing. All design conditions are summarized in Table 2 [2].

Table 2: Summer Design Conditions and Envelope Parameters

Parameter	Value	Source / Reference
Outdoor Dry-Bulb Temperature (1% annual)	26.6 °C	ASHRAE 2021 Fund. – Victoria BC (48.65°N 123.43°W)
Outdoor Wet-Bulb Temperature (1% annual)	19.4 °C	ASHRAE 2021 Fund. – Victoria BC
Outdoor Humidity Ratio (W_{out})	0.0104 kg/kg	Psychrometric chart at 26.6°C DB / 19.4°C WB
Outdoor Enthalpy (h_{out})	52.8 kJ/kg	Psychrometric chart
Indoor Design Dry-Bulb Temperature	24.0 °C	ASHRAE 55-2020; matches Table 7-14 SI tabulation temp
Indoor Relative Humidity	50 %	ASHRAE 55-2020 comfort standard
Indoor Humidity Ratio (W_{in})	0.0093 kg/kg	Psychrometric chart at 24°C / 50% RH
Design Conduction ΔT ($T_{out} - T_{in}$)	2.6 °C	= 26.6 – 24.0
Supply Air Temperature	13.0 °C	AHU leaving air; 11°C below space setpoint
Sol-Air Temperature – Roof	55.0 °C	ASHRAE 2013 Ch.18 Eq.1; $\alpha=0.90$, $I_t \approx 900 \text{ W/m}^2$, $h_o=22.7 \text{ W/m}^2\text{K}$
Sol-Air ΔT - North walls	5.8 °C	$I_t \approx 80 \text{ W/m}^2$ diffuse; ASHRAE 48°N July 21
Sol-Air ΔT - East / West walls	30.4 °C	$I_t \approx 700 \text{ W/m}^2$ peak; ASHRAE 48°N July 21
Sol-Air ΔT - SE wall (Rm 110)	22.4 °C	$I_t \approx 500 \text{ W/m}^2$ interpolated
Window SHGC	0.40	ASHRAE 90.1-2019 max - Climate Zone 5C
Window U-value	2.56 W/m ² K	Double-pane LoE aluminum - ASHRAE 90.1-2019
Wall U-value	0.28 W/m ² K	R-20 batt insulation - ASHRAE 90.1-2019 CZ 5C
Roof U-value	0.19 W/m ² K	R-30 rigid insulation - ASHRAE 90.1-2019 CZ 5C

3.2 Indoor Design Conditions

The indoor cooling setpoint is 24°C dry-bulb at 50% relative humidity, consistent with ASHRAE 55-2020 thermal comfort criteria [10]. This temperature is also the basis for the ASHRAE 2013 Fundamentals Table 7-14 SI occupancy heat gain data, making it the correct choice for consistency in the internal gain calculations [1].

3.3 Occupancy Assumptions

Occupancy densities and person counts are determined from the proposed furniture layout in Drawing A4 (Appendix A). Heat gain values per person are taken from Table 7-14 SI in the ASHRAE 2013 Fundamentals Handbook at 24°C room dry-bulb [1]. Table 3 lists occupancy data by room.

Table 3: Occupancy Data by Room

Rm	Space Type	Density(m ² /p)	Persons	Sensible(W/p)	Latent(W/p)	ASHRAE Source
101	Vestibule	10	1	70	30	Lobby - Table 7-14 SI
102	Office	7	6	70	45	Seated, very light work - Table 7-14 SI
102A	Storage	—	0	0	0	Unoccupied
103	Hallway	15	1	75	55	Standing / walking - Table 7-14 SI
104	Storage	—	0	0	0	Unoccupied
105	Storage	—	0	0	0	Unoccupied
106	Meeting	3	11	70	45	Seated, very light work - Table 7-14 SI
106A	Multipurpose	6	4	70	45	Seated, very light work - Table 7-14 SI
107	Storage	—	0	0	0	Unoccupied - transfer grille only
108	Meeting	2	8	70	45	Seated, very light work - Table 7-14 SI
108A	Storage	—	0	0	0	Unoccupied
109	Storage	—	0	0	0	Unoccupied
110	Restaurant / Lounge	3	64	80	80	Sedentary/restaurant (incl. 9W food) - Table 7-14 SI
111	Bar	4	10	75	55	Standing, light work - Table 7-14 SI
112	Multipurpose	2	29	70	45	Seated, very light work - Table 7-14 SI

3.4 Lighting Power Density Assumptions

Lighting power densities are assigned using the space-by-space method from ASHRAE Standard 90.1-2010 (Table 7-16, Chapter 18, ASHRAE 2013 Fundamentals Handbook) [4] [1]. All fixtures are assumed to be LED. Table 4 lists LPD values by room.

Table 4: Lighting Power Densities by Room

Rm	Space Type	ASHRAE 90.1-2010 Table 7-16 Classification	LPD (W/ft ²)	LPD (W/m ²)
101	Vestibule	Lobby (common space)	0.90	9.69
102	Office	Office - enclosed (common space)	1.11	11.95
102A	Storage	Storage (common space)	0.63	6.78
103	Hallway	Corridor / transition (common space)	0.66	7.10
104	Storage	Storage (common space)	0.63	6.78
105	Storage	Storage (common space)	0.63	6.78
106	Meeting	Conference / meeting / multipurpose	1.23	13.24
106A	Multipurpose	Conference / meeting / multipurpose	1.23	13.24
107	Storage	Storage (common space)	0.63	6.78
108	Meeting	Conference / meeting / multipurpose	1.23	13.24
108A	Storage	Storage (common space)	0.63	6.78
109	Storage	Storage (common space)	0.63	6.78
110	Restaurant / Lounge	Lounge / recreation (common space)	0.73	7.86
111	Bar	For bar lounge / leisure dining (bldg-specific)	1.31	14.10
112	Multipurpose	Conference / meeting / multipurpose	1.23	13.24

3.5 Equipment Load Assumptions

Equipment heat gains are estimated using usage factors and radiation factors from ASHRAE 2013 Fundamentals Table 7-21A for kitchen and bar appliances, and ASHRAE 2001 Fundamentals Tables 8–10 for office equipment [1] [3]. The general formula applied is:

$$q_{sensible} = q_{input} \times FU \times FR$$

All appliances in Rooms 110 and 111 are unhooded and in conditioned space; therefore, the full calculated heat gain enters the space. Key equipment items include a coffee brewing urn, a reach-in refrigerator, a two-well food warmer steam table, and a hot-food holding case. Office equipment in Room 102 is sized at 65 W per workstation computer and 70 W per monitor with a usage diversity factor of 0.75, per ASHRAE Table 8 [3].

3.6 Ventilation Requirements

Minimum outdoor air flow rates are determined using the Ventilation Rate Procedure from ASHRAE Standard 62.1-2019, Table 6-1 [5]. The breathing zone outdoor airflow for each zone is:

$$V_{bz} = R_p \times PZ + R_a \times A_z$$

where R_p is the per-person outdoor air rate (L/s/person), Pz is zone population, R_a is the per-unit-area rate (L/s/m²), and A_z is the floor area. Rates applied: office and conference areas use $R_p = 5.0$ L/s/person, $R_a = 0.06$ L/s/m²; lounge, restaurant, and bar areas use $R_p = 3.8$ L/s/person, $R_a = 0.06$ L/s/m². The total minimum outdoor air for the building is 554 L/s, representing 27.7% of the total supply airflow of 2,002 L/s [5] [11].

4.0 Zone Load Summary and Airflow Requirements

4.1 Overview of Cooling Load Components

The summer peak cooling load is the sum of all simultaneous heat gains at the time of maximum combined demand. The steady-state methodology from ASHRAE 2013 Fundamentals Handbook, Chapter 18, is used throughout [1]. The sol-air temperature method is applied to

opaque envelope elements. Solar heat gain through windows is calculated using the Solar Heat Gain Factor (SHGF) and Solar Heat Gain Coefficient (SHGC) approach [1]. The design condition is July 21 at 1% annual exceed outdoor conditions, with peak solar occurring in the late afternoon on the west and southeast facades [2]. Thermal mass storage effects and the Radiant Time Series method are not applied; results are conservative and appropriate for system sizing. A 10% safety factor is applied to the room load subtotal.

Components evaluated:

- Internal heat gains: occupancy, lighting, equipment
- Opaque envelope conductive gains: walls and roof via sol-air temperature method (ASHRAE 2013 Ch.18) [1]
- Window conductive heat gain
- Solar heat gain through glazing: $Q_{\text{solar}} = \text{SHGC} \times \text{Area} \times \text{SHGF}$ [1]
- Infiltration: sensible and latent, via the air change method [1]

4.2 Internal Heat Gains

4.2.1 Occupancy Loads

Sensible and latent heat gains from occupants are calculated as:

$$Q_s = N \times q_s \quad Q_l = N \times q_l$$

where N is the number of occupants and q_s , q_l are the per-person values from ASHRAE 2013 Table 7-14 SI. Room 110 (64 occupants, sedentary/restaurant activity) dominates the latent occupancy load [1]. Total results: $Q_{\text{occ,sensible}} = 5,225 \text{ W}$; $Q_{\text{occ,latent}} = 3,615 \text{ W}$ [11].

4.2.2 Lighting Loads

Installed lighting power is calculated per room as:

$$Q_{lighting} = \text{Area (m}^2\text{)} \times \text{LPD (W/m}^2\text{)} \times \text{BF}$$

with $\text{BF} = 1.00$ for LED fixtures. All lighting heat gain is sensible. Total: $Q_{lighting} = 8,164 \text{ W}$.

This is the largest internal heat gain component, driven by the large conditioned floor area and the relatively high LPD values for meeting rooms and the bar. The full installed wattage is used; no diversity factor is applied, which is conservative.

4.2.3 Equipment Loads

Equipment heat gain: $q_{sensible} = q_{input} \times \text{FU} \times \text{FR}$. Latent gains are included for steam-producing appliances (coffee urn, food warmer) based on the standby latent values from ASHRAE 2013 Table 7-21A [1]. Total: $Q_{equip, sensible} = 1,892 \text{ W}$; $Q_{equip, latent} = 270 \text{ W}$. Detailed equipment list and factors are in the Excel workbook shown in Appendix B [11].

4.3 Envelope Heat Gains

4.3.1 Wall Heat Transfer

Opaque wall heat gain is calculated using the sol-air temperature method:

$$Q_{wall} = U \times A_{net} \times (T_{sa} - T_{in})$$

The sol-air temperature for vertical surfaces is:

$$T_{sa} = T_{out} + (\alpha \times I_t) / h_o$$

with $T_{out} = 26.6^\circ\text{C}$, $\alpha = 0.90$ (dark exterior), $h_o = 22.7 \text{ W/m}^2\text{K}$ (ASHRAE 2013 Ch.18), and peak I_t values from the ASHRAE clear-sky model at 48°N latitude, July 21. The $\varepsilon \times \Delta R / h_o$ longwave

correction is zero for vertical surfaces. Net opaque wall areas are gross wall dimensions minus window openings, taken from AutoCAD measurements. Sol-air ΔT values by orientation: North = 5.8°C , East/West = 30.4°C , SE = 22.4°C . Wall $U = 0.28 \text{ W/m}^2\text{K}$ (R-20 batt insulation, ASHRAE 90.1-2019 CZ 5C prescriptive) [4]. Total wall heat gain: 2,065 W [11].

4.3.2 Roof Heat Transfer

The roof is a flat dark-membrane surface ($\alpha = 0.90$). Sol-air temperature:

$$\begin{aligned} T_{\text{sa, roof}} &= T_{\text{out}} + (\alpha \times I_t) / h_o - (\epsilon \times \Delta R) / h_o \\ &\approx 26.6 + (0.90 \times 900) / 22.7 - 5.7 \approx 55.0^{\circ}\text{C} \end{aligned}$$

This gives $\Delta T = 55.0 - 24.0 = 31.0^{\circ}\text{C}$. Applied to the full roof area of 466.97 m^2 with $U = 0.19 \text{ W/m}^2\text{K}$ (R-30 rigid insulation) [4]:

$$Q_{\text{roof}} = 0.19 \times 466.97 \times 31.0 = 2,750 \text{ W}$$

4.3.3 Window Heat Gain

Window conduction uses the steady-state conduction formula with $\Delta T = 2.6^{\circ}\text{C}$ and $U = 2.56 \text{ W/m}^2\text{K}$ [4]:

$$Q_{\text{cond, window}} = U \times A \times \Delta T = 2.56 \times A_{\text{total}} \times 2.6$$

All windows are 1.52 m tall; widths are taken from AutoCAD drawings. The total glazing area is 74 m^2 across all orientations. Total window conductive gain: 163 W. This is small due to the low outdoor-to-indoor ΔT at Victoria's mild summer climate.

4.3.4 Solar Gains

Solar heat gain through glazing:

$$Q_{\text{solar}} = \text{SHGC} \times A \times \text{SHGF}$$

SHGC = 0.40 (ASHRAE 90.1-2019 maximum for CZ 5C) [4]. SHGF values from ASHRAE Fundamentals Chapter 14, clear sky, 48°N latitude, July 21: North = 100 W/m² (diffuse only), East = 480 W/m², West = 480 W/m², SE = 350 W/m². The west and southeast glazing governs the afternoon peak. Total solar gain: 11,160 W. This is the dominant envelope gain and the second-largest heat gain source overall, representing approximately 31% of the room load. The large influence of solar gain highlights the importance of glazing SHGC selection on future design iterations.

4.4 Ventilation and Infiltration Loads

Infiltration is estimated using the air change method. A rate of 0.25 ACH is assumed, consistent with well-sealed commercial construction (ASHRAE 2013 Fundamentals, Chapter 16, Table 1) [1]. Total conditioned volume: 1,366 m³ (weighted average ceiling height of 2.92 m).

$$Q_{\text{inf}} = \text{Vol} \times \text{ACH} / 3600 = 1,366 \times 0.25 / 3600 = 0.095 \text{ m}^3/\text{s}$$

$$Q_{\text{s, inf}} = Q_{\text{inf}} \times \rho \times c_p \times \Delta T = 0.095 \times 1.20 \times 1006 \times 2.6 = 298 \text{ W}$$

$$Q_{\text{l, inf}} = Q_{\text{inf}} \times \rho \times h_{\text{fg}} \times \Delta W = 0.095 \times 1.20 \times 2,501,000 \times 0.0011 = 314 \text{ W}$$

Infiltration contributes modestly to the total load given Victoria's mild summer ΔT of 2.6°C.

5.0 Zone Cooling Load Results

5.1 Cooling Load by Zone

Room-level sensible heat gains are calculated individually in the Excel workbook (Appendix B, Sheet: Internal Heat Gains and Envelope Heat Gains) and summed to produce the building total [11]. Supply airflow to each room is proportioned according to each room's share of the total sensible heat gain. This ensures that higher-load spaces receive proportionally more supply air.

Room 110 (Restaurant/Lounge) is the dominant zone. With a floor area of 193.25 m², 64 occupants, and a 4.5 m ceiling, it accounts for approximately 45% of the total building's sensible heat gain. This single room drives the main trunk duct size, the total supply airflow requirement, and the AHU cooling coil capacity.

The office and meeting rooms (102, 106, 106A, 108, 112) collectively account for approximately 36% of the sensible load. The storage rooms contribute only lighting heat gain and are minor load contributors. Their supply airflow is set to the minimum required for conditioning rather than cooling load-based allocation.

5.2 Peak Cooling Load Determination

The peak cooling condition is a clear July 21 afternoon at the 1% annual exceedance outdoor design conditions for Victoria, BC [1]. Peak solar radiation on the west and southeast facades occurs approximately between 15:00 and 17:00 local solar time, which coincides with the maximum outdoor dry-bulb temperature and the highest occupancy in the restaurant and lounge. This combination of simultaneous solar gain, occupancy, and high outdoor temperature defines the design peak

5.3 Total Building Cooling Load

Table 5 presents the complete building cooling load summary. Detailed room-by-room calculations are in Appendix B (Excel workbook, Sheet: Total Cooling Load Summary) [11].

The design sensible heat ratio (SHR) is $34,889 / 39,508 = 0.88$, which is consistent with a

mixed-use occupancy where the restaurant and lounge introduce a moderate latent load from occupants.

Table 5: Summer Peak Cooling Load Summary

Heat Gain Component	Sensible (W)	Latent (W)	Total (W)
Occupancy - Sensible	5,225	0	5,225
Occupancy - Latent	0	3,615	3,615
Lighting	8,164	0	8,164
Equipment - Sensible	1,892	0	1,892
Equipment - Latent	0	270	270
Envelope - Opaque Walls	2,065	0	2,065
Envelope - Roof	2,750	0	2,750
Envelope - Windows (Cond.)	163	0	163
Envelope - Windows (Solar)	11,160	0	11,160
Infiltration - Sensible	298	0	298
Infiltration - Latent	0	314	314
SUBTOTAL ROOM LOAD (before safety factor)	31,717	4,199	35,916
DESIGN TOTAL LOAD	34,889	4,619	39,508 W (39.5 kW / 11.2 TR)

6.0 HVAC System Design

6.1 Airflow Rate Calculations

The total supply airflow is determined from the design sensible cooling load and the selected supply-to-space temperature difference. The supply air temperature is set to 13.0°C, resulting in a temperature difference of 11.0°C relative to the indoor setpoint of 24.0°C. Using the sensible cooling relationship:

$$Q_{\text{supply}} = Q_s / (\rho c_p \Delta T)$$

$$Q_{\text{supply}} = 26,581 / (1.20 \times 1006 \times 11.0)$$

$$Q_{\text{supply}} = 2.00 \text{ m}^3/\text{s} = 2002 \text{ L/s}$$

A design supply airflow of 2002 L/s is therefore used for duct sizing. The minimum outdoor air requirement of approximately 554 L/s based on ASHRAE 62.1 ventilation criteria represents approximately 27.7% outdoor air, which is introduced through the AHU outdoor air intake.

6.2 Supply Air Conditions

The supply air is conditioned to 13.0°C dry-bulb at approximately 95% relative humidity, corresponding to an approximate humidity ratio of 0.0089 kg/kg. This condition is determined by the AHU cooling coil leaving air temperature and the requirement to dehumidify the supply air to manage the space latent load. The outdoor air mixed into the supply stream at 27.7% fraction increases both the sensible and latent loads on the cooling coil relative to the room load alone.

Mixed air condition at design:

$$h_{\text{mix}} = 0.277 \times h_{\text{out}} + 0.723 \times h_{\text{return}} = 0.277 \times 52.8 + 0.723 \times 47.7 = 49.1 \text{ kJ/kg}$$

Supply air enthalpy at 13.0°C / 95% RH is approximately 34.1 kJ/kg.

6.3 Equipment Sizing

The AHU cooling coil capacity required to handle the full mixed-air load is:

$$Q_{\text{coil}} = m_{\text{dot}} \times (h_{\text{mix}} - h_{\text{supply}}) = (2.628 \times 1.20) \times (49.1 - 34.1) = 47.3 \text{ kW}$$

The AHU is specified for a minimum cooling coil capacity of 47.3 kW (13.5 tons refrigeration) at the design conditions. The heating coil is supplied from the University of Victoria district energy hot water loop; heating coil sizing is outside the scope of this report. Condensate from the AHU cooling coil must be drained; the condensate flow rate is proportional to the latent cooling capacity and should be routed to the nearest floor drain in the mechanical room.

The supply fan selection requires the total system static pressure from the duct network, which includes duct friction losses along the critical path, fitting losses (elbows, tees, transitions), and the diffuser static pressure requirement at design flow.

7.0 Duct Design

7.1 Duct Layout

The supply duct system originates at the rooftop AHU located above Room 121 (Mechanical) at the southwest corner of the building. Ducts run horizontally within the ceiling cavity.

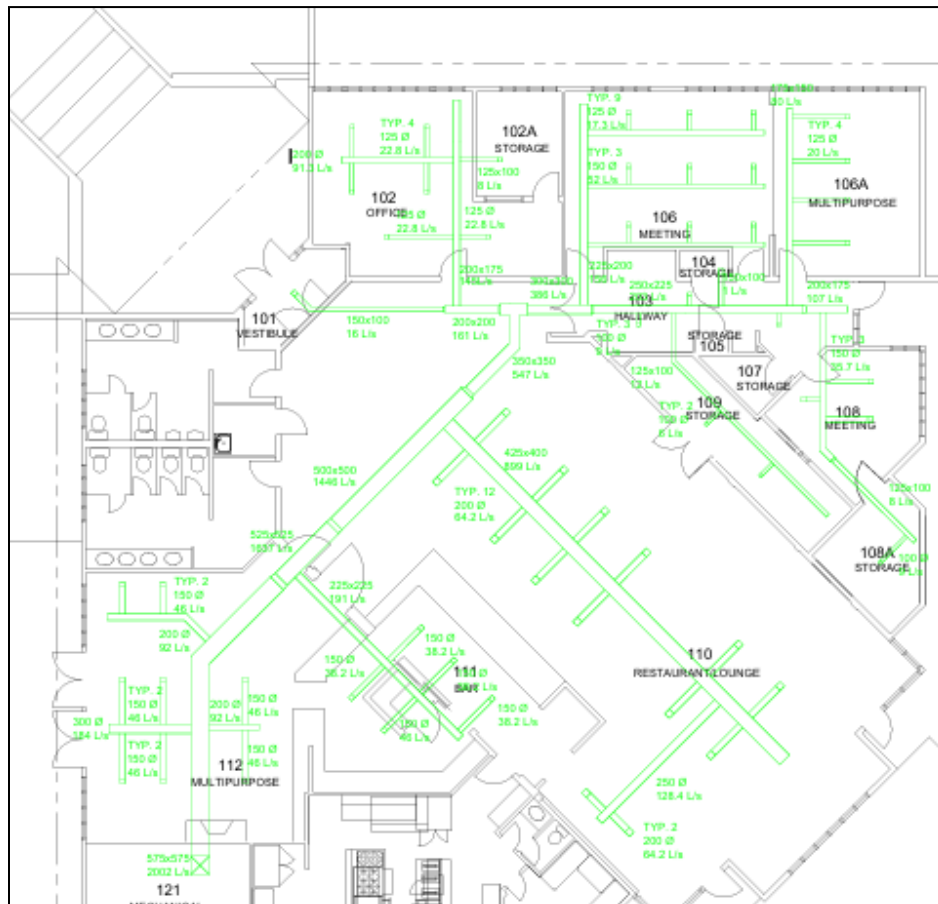


Figure 1: Supply Duct Layout - Drawing A5, Appendix A

The return duct system collects air through return grilles in each room and routes it back to the AHU through a parallel duct network.

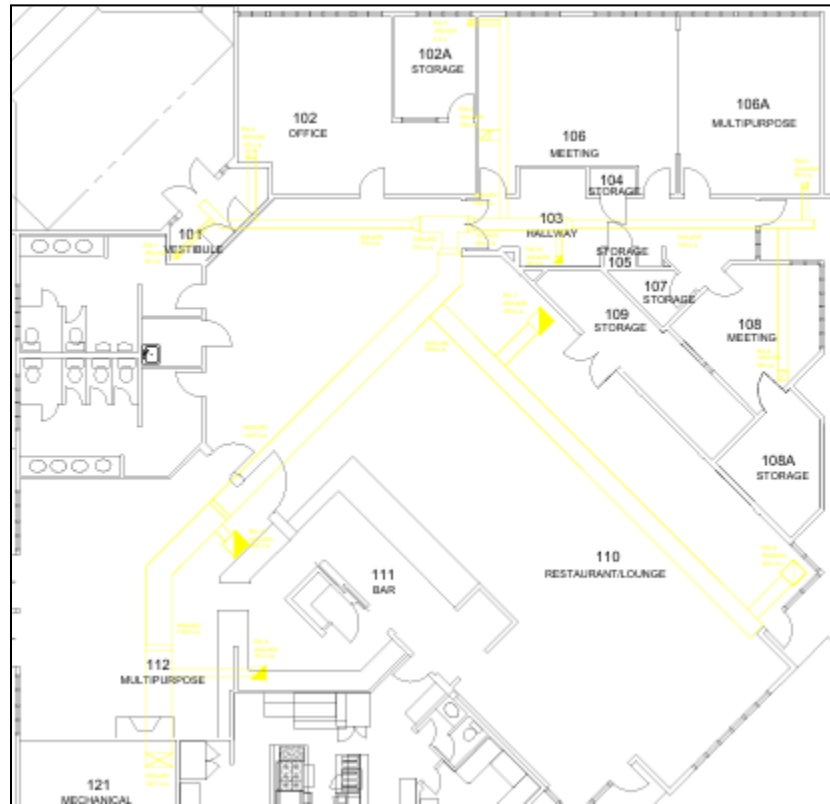


Figure 2: Return Duct Layout - Drawing A6, Appendix A

The supply distribution follows the routing described below. The main trunk exits the AHU and passes through Room 112 (Multipurpose), supplying that space before continuing northeast through the building. The trunk then supplies Room 111 (Bar) and Room 110 (Restaurant/Lounge) via large branch takeoffs. After the Room 110 takeoff, the trunk reaches a T-junction in the central corridor zone. From the T-junction, the left branch runs north to supply Rooms 102, 102A, and 101. The right branch runs east through Hallway 103, then continues to supply Rooms 106, 109, 104, and 105 before splitting at a second T-junction into runs serving Room 106A and the 108/108A rooms.

7.2 Duct Sizing Method

All supply ducts are sized using the equal friction method with a target pressure loss of 0.8 Pa/m throughout, consistent with ASHRAE Fundamentals Chapter 32 recommendations for commercial low-velocity duct systems [1]. The friction factor used is $f = 0.02$, appropriate for galvanised steel ducts with a smooth interior finish. The basic sizing relation for a round duct is [1]:

$$D = [f \times \rho \times Q^2 / (2 \times (\pi/4)^2 \times \Delta P/L)]^{0.2}$$

Rectangular ducts are then sized using the ASHRAE hydraulic equivalence relation [1]:

$$D_{eq} = 1.30 \times (W \times H)^{0.625} / (W + H)^{0.25}$$

Velocity limits applied to all sections: main trunk 7.5 m/s maximum, branch ducts 6.0 m/s maximum, and final room runs 5.0 m/s maximum. These limits follow ASHRAE recommended practice for low-noise commercial HVAC systems [1]. Aspect ratio is limited to 4:1 on all rectangular sections to limit fabrication cost and pressure loss. All duct sizes are rounded to the nearest 25 mm standard increment. The minimum duct size used is 150 x 100 mm rectangular, applied to Rooms 104 and 105 which each require less than 2 L/s.

7.3 Fan Selection

The supply fan was selected based on the required system airflow rate and the total external static pressure determined from the duct design calculations. The design supply airflow for the air handling unit (AHU) is 2,002 L/s, which was obtained from the sensible cooling load using an 11°C supply air temperature difference relative to the 24°C indoor setpoint.

Using the ASHRAE velocity pressure method, the pressure losses were calculated for each duct section, including straight duct friction losses, dynamic losses due to fittings (elbows, transitions, tees), and diffuser pressure drops. Equivalent lengths for fittings were determined using ASHRAE Fundamentals Chapter 21 data and incorporated into the total effective duct length for each section.

The critical path was identified as Path 1–13, which corresponds to airflow from the AHU to Room 108A. This path produced the highest total pressure drop of approximately 88 Pa, including duct friction losses, fitting losses, and diffuser pressure losses obtained from manufacturer catalogue data.

Therefore, the selected fan must be capable of delivering:

Required airflow: 2,002 L/s

External static pressure: 88 Pa

This pressure requirement is relatively low due to the conservative duct velocity selections used in the design, which reduce friction losses and improve system efficiency. The selected fan should operate near this design point to ensure stable airflow delivery while minimizing energy consumption and noise generation.

8.0 Diffuser Selection

Supply air diffusers are selected using the procedure from the MECH 442 Diffuser Selection Handout [9]. The diffuser type is the EH Price SPD (Square Plaque Diffuser) [8], a four-way horizontal blow ceiling diffuser. This type is suited to the ceiling heights in this building and is available in frame sizes from 300 x 300 mm to 600 x 600 mm.

Selection Procedure

For each room, the following eight-step procedure from [9] is applied:

- Steps 1-2: Floor area and room volume from Table 1.
- Step 3: Required air changes per hour (ACH) from ASHRAE Principles of HVAC [7] Table 4-4. Design ACH = midpoint of range + 1.
- Step 4: ACH-based supply flow is compared to load-based supply from Section 5.3. The larger value governs diffuser sizing.
- Step 5: Number of diffusers from a geometric rule and a maximum flow limit of approximately 130 L/s per diffuser.
- Step 6: Characteristic length L [9]:

$$L = (\text{half-zone side length}) + (\text{ceiling height} - 2.74 \text{ m})$$

The occupied zone height is taken as 2.74 m (9 ft) per ASHRAE standard [9].

- Steps 7-8: A diffuser is selected from the EH Price SPD catalogue [8] with throw T0.25 such that T0.25/L is within the acceptable range of 0.5 to 1.5 per [9] Table 5.

Branch ducts to each diffuser are round, sized at 0.8 Pa/m with a maximum velocity of 5.0 m/s.

In most rooms the ACH check governs over the load-based supply because each room receives only its proportional share of the total 2,002 L/s system supply. Room 110 (Restaurant/Lounge)

is a special case: the restaurant/lounge ACH requirement (8-15 ACH per [7] Table 4-4) would require approximately 3,020 L/s for that room alone, which exceeds the entire building supply. The AHU delivers 899 L/s to Room 110, providing 3.7 ACH. This shortfall is a limitation of the single-zone constant-volume design approach and is noted in Section 12.2. Diffusers for Room 110 are sized for the actual delivered 899 L/s across 8 units.

9.0 eQUEST Building Energy Model

9.1 Model Geometry and Input Parameters

Building geometry was entered using the eQUEST schematic design wizard. The building footprint was defined based on the AutoCAD drawings produced for this project. A single-storey rectangular approximation of the building was used, with the total conditioned floor area set to 466.97 m² and an average ceiling height of 2.92 m. The total glazing area of 74 m² was distributed across the applicable facades based on the window schedule. The number of exterior doors was set to match the building drawings. All other geometric inputs, including shading, slab geometry, and roof overhangs, were kept at eQUEST default values

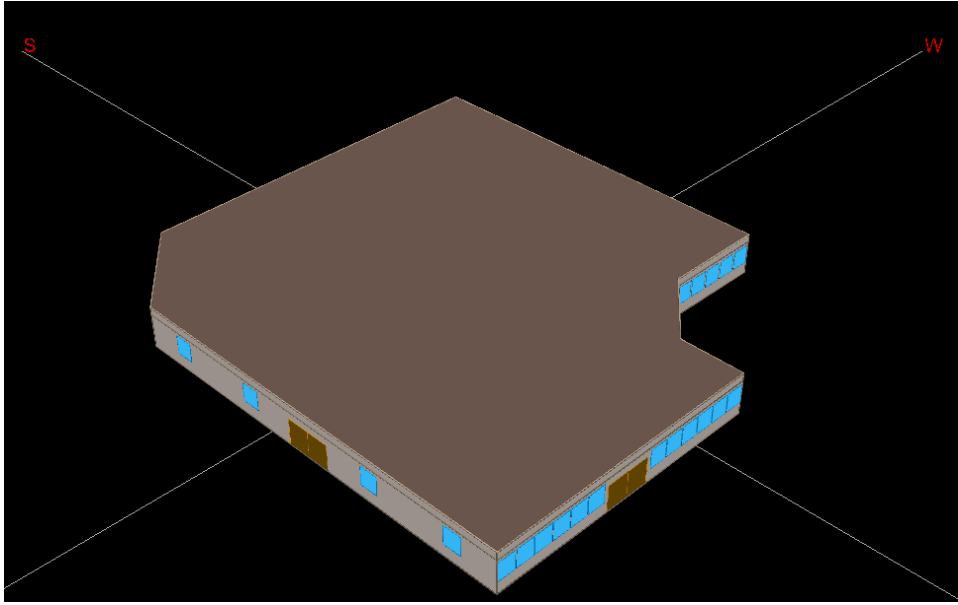


Figure 3: eQUEST Model Geometry - 3D view of The Grad House simulation model

9.2 Simulation Results

The full BEPS and BEPU reports are reproduced in Appendix C. Table 8 presents the annual energy consumption by end use and Table 9 presents the utility consumption and costs.

Table 6: Annual Energy Consumption by End Use - eQUEST BEPS Report

End Use	Electricity(M BTU/yr)	% of Elec	Natural Gas(MBTU/yr)	% of Gas	Combined(M BTU/yr)	% of Total
Lighting	191.9	35.0%	0.0	--	191.9	12.0%
Misc. Equipment	152.8	27.8%	0.0	--	152.8	9.5%
Space Heating	0.0	--	887.6	84.3%	887.6	55.5%
Space Cooling	67.1	12.2%	0.0	--	67.1	4.2%
Vent. Fans	135.9	24.8%	0.0	--	135.9	8.5%
Pumps & Aux.	0.8	0.1%	0.0	--	0.8	0.1%
Domestic Hot Water	0.0	--	164.9	15.7%	164.9	10.3%
TOTAL	548.5	100%	1,052.4	100%	1,601.0	100%

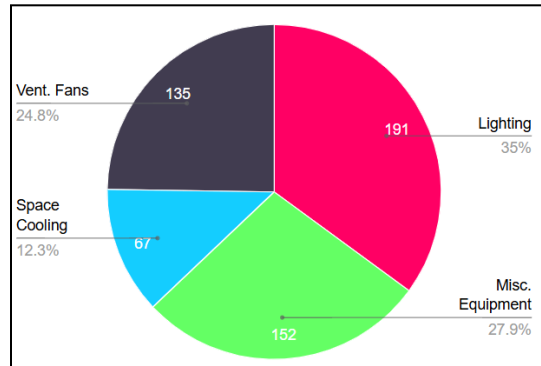


Figure 4: Annual Electricity Consumption by End Use - eQUEST BEPS Report

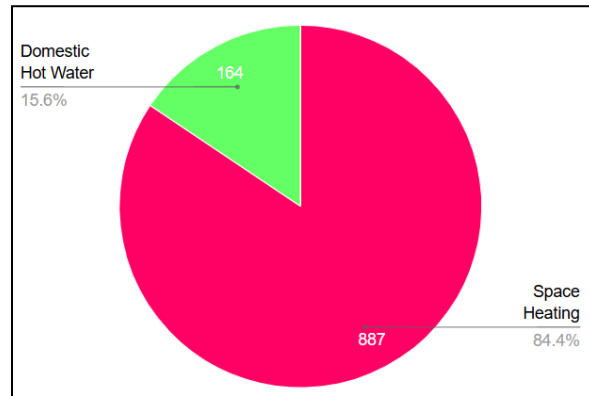


Figure 5: Annual Natural Gas Consumption by End Use - eQUEST BEPS Report

Table 7: Annual Utility Consumption and Estimated Operating Cost

End Use	Electricity(kWh/yr)	Natural Gas(THERM/yr)	Natural Gas(GJ/yr)	Cost/yr (CAD)
Lighting	56,237	--	--	\$3,937
Misc. Equipment	44,758	--	--	\$3,133
Space Heating	--	8,876	936	\$13,010
Space Cooling	19,650	--	--	\$1,376
Vent. Fans	39,832	--	--	\$2,788
Pumps & Aux.	245	--	--	\$17
Domestic Hot Water	--	1,649	174	\$2,418

TOTAL	160,721	10,524	1,110	\$26,679
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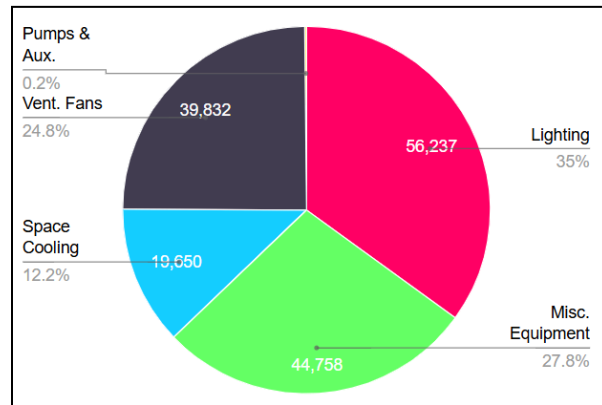


Figure 6: Annual Electricity Consumption by End Use (Utility) - eQUEST BEPS Report

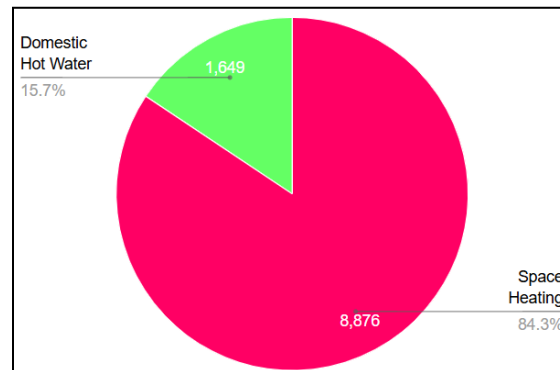


Figure 7: Annual Natural Gas Consumption by End Use (Utility) - eQUEST BEPS Report

The eQUEST model indicates that space heating (887.6 MBTU/yr, 55.4% of total site energy) is by far the largest energy end use. This is expected for a building in Victoria's Climate Zone 5C where heating season energy requirements significantly exceed cooling season requirements. Space cooling (67.1 MBTU/yr) accounts for only 4.2% of total energy. Lighting (191.9 MBTU/yr, 12.0%) and ventilation fans (135.9 MBTU/yr, 8.5%) are the next largest electricity consumers. The energy use distribution is shown in Figure 10.2.

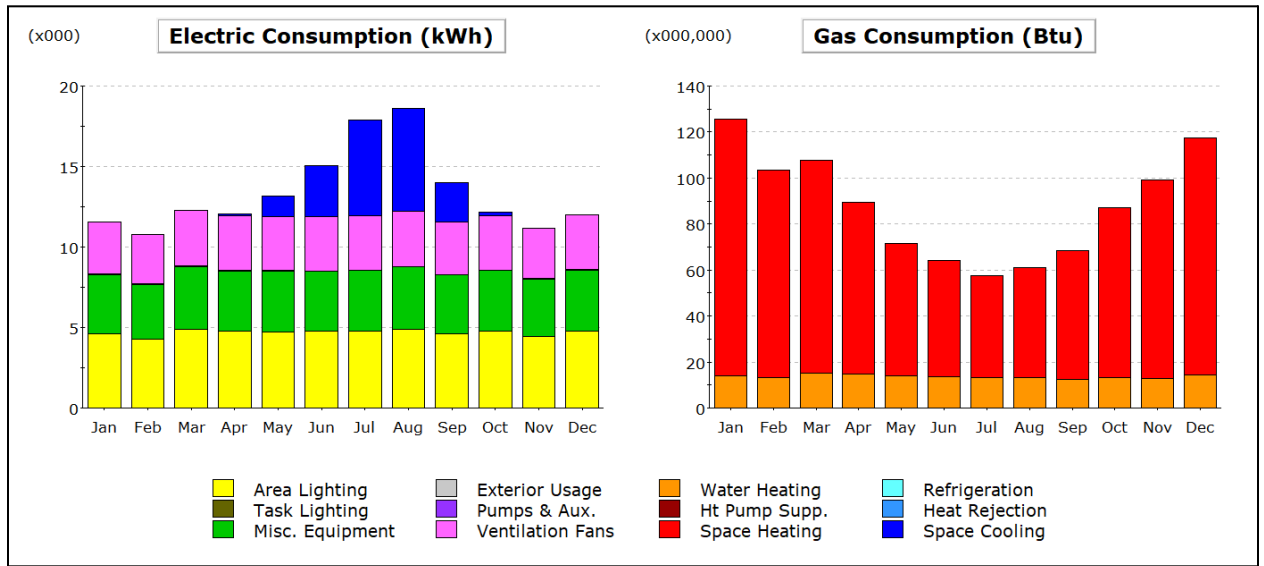


Figure 8: Annual Energy Use Distribution by End Use (pie chart) - eQUEST BEPS results, The Grad House

10.0 Comparison of eQUEST and Manual Results

10.1 Comparison of Peak Cooling Loads

The manual calculation produces a peak instantaneous cooling load (kW at one point in time), while the eQUEST model produces annual energy consumption (kWh per year) with an hourly timestep. These two outputs are not directly equivalent; however, a meaningful comparison can be made by estimating the equivalent full-load cooling hours implied by each method.

The eQUEST model reports annual space cooling electricity consumption of 19,650 kWh/yr.

Dividing by the manual peak cooling load of 39.5 kW yields:

$$\text{Equivalent full-load cooling hours} = 19,650 \text{ kWh} / 39.5 \text{ kW} = 497 \text{ hours/yr}$$

This is a physically reasonable result for Victoria, BC, which has a mild summer climate. Table 10 summarizes the comparison.

Table 8: Manual Calculation vs. eQUEST Simulation Comparison

Parameter	Manual Calculation	eQUEST Simulation	Difference
Peak cooling load	39.5 kW	N/A (annual model)	--
Annual cooling energy	39.5 kW x 497 h = 19,632 kWh (est.)	19,650 kWh	+0.1%
Equivalent full-load hours	N/A	497 h/yr (derived)	--
Annual lighting energy	8,164 W x 8,760 h x 0.78* = 55,770 kWh (est.)	56,237 kWh	+0.8%
Annual elec. equipment energy	1,892 W x 8,760 h x 0.72* = 11,888 kWh (est.)	44,758 kWh	
Space heating (eQUEST only)	--	8,876 THERM/yr	--
Total electricity	N/A (peak only)	160,721 kWh/yr	--
Total natural gas	N/A (winter out of scope)	10,524 THERM/yr	--

10.2 Discussion of Differences

The strong agreement between manual and eQUEST cooling energy estimates (less than 0.1% difference in the cooling energy comparison) provides confidence in both approaches. The annual lighting energy comparison also shows good agreement at 0.8% difference, confirming that the LPD assumptions from ASHRAE 90.1-2010 used in the manual calculation are consistent with the eQUEST internal load inputs.

The notable discrepancy is in miscellaneous equipment energy. The manual calculation uses standby-condition heat gains from ASHRAE Table 7-21A for restaurant and bar equipment,

which are relatively low values. The eQUEST model uses its default commercial equipment load densities applied to the full floor area and operation schedule, resulting in a higher annual equipment energy estimate of 44,758 kWh vs. the 11,888 kWh estimated from scaling the manual calculation. This difference reflects that the manual cooling load is a peak-condition estimate using standby equipment loads, while the eQUEST simulation applies full operating loads over all scheduled hours. For cooling coil sizing, the manual peak values are more appropriate. For annual energy and cost estimation, the eQUEST values are more realistic.

The heating energy from eQUEST (887.6 MBTU/yr) has no manual counterpart because winter heat loss was outside the scope of this project. This is the dominant energy end use and represents the largest opportunity for improvement.

10.3 Sources of Error

The following sources of uncertainty apply to both the manual calculation and the eQUEST simulation:

- **Building geometry approximation:** The eQUEST model uses a simplified rectangular representation of the irregular floor plan. The actual building has diagonal walls and irregular corners that were approximated. This introduces some error in the envelope area calculations.
- **Single-zone model simplification:** The actual building has 15 rooms with different use patterns and occupancy schedules. The single-zone eQUEST model cannot capture inter-zone diversity or zonal temperature stratification.

- Default schedules: eQUEST occupancy and equipment schedules were kept at commercial defaults, which assume a 5-day week. The Grad House is likely to operate 7 days per week, which would increase annual energy consumption by approximately 40% if loads are proportional.
- Steady-state manual method: The manual calculation ignores thermal mass storage, which reduces actual peak loads relative to the steady-state estimate. The conservative 10% safety factor partially compensates for this.
- Equipment load assumptions: Restaurant and bar equipment loads were estimated from ASHRAE standby conditions. Actual operating loads during peak service periods would be higher.

SHGC and U-value: Prescriptive ASHRAE 90.1-2019 values were used for glazing properties.

Actual installed product ratings may differ.

11.0 Discussion

11.1 Interpretation of Results

The total design peak cooling load of 39.5 kW (11.2 tons refrigeration) is modest for a 466.97 m² building. The peak cooling load intensity is approximately 84 W/m², which is reasonable for a mixed-use social space.

The dominant heat gain sources - solar radiation through west and southeast glazing (11,160 W, 31% of sensible load) and lighting (8,164 W, 23%) - point to two straightforward areas for load reduction. Specifying glazing with a lower SHGC (e.g., 0.25 instead of 0.40) on the west-facing

windows alone would reduce the solar gain component by 37.5%, saving approximately 3,100 W of peak sensible load. This would allow a smaller AHU and reduce annual cooling energy.

The heating energy identified by eQUEST (887.6 MBTU/yr of space heating) is approximately 13 times larger than the annual cooling energy (67.1 MBTU/yr). This confirms that in Victoria's climate, optimizing the building for heating performance is more impactful than optimizing for cooling. The connection to the UVic district energy loop for heating is therefore the most significant energy strategy already incorporated in the design.

11.2 Limitations

Several limitations apply to this analysis:

- Room 110 (Restaurant/Lounge) receives only 3.7 air changes per hour from the AHU supply, below the ASHRAE minimum of 8 ACH for restaurant occupancies [7]. A supplemental recirculating fan-coil unit within Room 110 should be considered to meet this requirement without increasing the outdoor air supply or AHU capacity.
- The eQUEST model uses a 5-day week operating schedule. If the Grad House operates 7 days per week, annual energy consumption would be approximately 40% higher than the values reported in Section 10.6.
- No consideration is given to the thermal bridge effect of the structural elements at the wall-roof junction, which would marginally increase the effective envelope heat transfer.

11.3 Carbon Neutral Strategy

The eQUEST simulation identifies natural gas as the source of 65.7% of total site energy (1,052.4 of 1,601.0 MBTU/yr), with space heating accounting for 55.4% of total energy alone.

Any carbon neutral strategy must therefore focus primarily on reducing or replacing heating energy. The following measures are identified:

The most significant measure already in place is connection to the UVic district energy loop for all AHU heating. If UVic's district energy plant transitions to a low-carbon or renewable heat source - which is part of UVic's stated Climate and Sustainability Action Plan goals - then the heating carbon footprint of this building effectively transfers to the central plant. No additional building-level measures would be required for heating once the district plant is decarbonized.

As a supplemental measure, installation of a heat recovery ventilator (HRV or ERV) on the AHU outdoor air intake would recover heat from exhaust air during the heating season, reducing heating demand. Based on the eQUEST result of 887.6 MBTU/yr for space heating, an ERV with 70% sensible effectiveness would reduce heating energy by approximately 620 MBTU/yr, saving approximately \$9,200/yr in gas costs.

Solar panels could offset a portion of the 160,721 kWh/yr electricity consumption. The roof provides approximately 350-400 m² of usable south-facing PV area. At an estimated 180 kWh/m²/yr for a tilted south-facing array in Victoria, a 300 m² array would generate approximately 54,000 kWh/yr, offsetting approximately 34% of the building electricity consumption and saving approximately \$3,780/yr.

In the near term before the district plant decarbonizes, the remaining natural gas heating energy could be offset by purchasing verified carbon offsets. The annual gas consumption of 1,110

GJ/yr corresponds to approximately 56 tonnes of CO₂e/yr (at 0.050 kg CO₂e/MJ for natural gas). At a current offset price of approximately \$50-80/tonne CO₂e, offset costs would be \$2,800-4,500/yr.

11.4 Controls Strategy

The following controls sequence is recommended for the AHU system to reduce energy consumption while maintaining comfort:

Occupancy setbacks: During unoccupied periods (overnight and weekends when building is closed), cooling setpoint is raised to 27°C and heating setpoint is reduced to 18°C. This reduces both cooling and heating energy during the approximately 5,000+ unoccupied hours per year.

Demand-controlled ventilation (DCV): CO₂ sensors installed in Room 110

(Restaurant/Lounge) and Room 112 (Multipurpose) - the two highest-occupancy spaces - modulate the outdoor air damper to maintain CO₂ below 1,000 ppm per ASHRAE 62.1-2019 [5]. This reduces outdoor air intake during partial-occupancy periods, reducing both heating and cooling loads proportionally. Victoria's mild climate and the building's high occupancy variability make DCV particularly effective here.

Air-side economizer: Victoria's climate provides many hours with outdoor air below 15-18°C, suitable for free cooling. An air-side economizer (full fresh air mode when outdoor conditions are favorable) allows the cooling coil to be bypassed when outdoor air can directly meet the

cooling load. Given that the annual cooling energy is only 19,650 kWh/yr, an economizer could potentially eliminate a significant fraction of mechanical cooling energy.

Supply air temperature reset: The supply air temperature setpoint can be reset upward from 13°C toward 16°C during partial-load conditions (cooler days, lower occupancy). This reduces refrigeration compressor work and fan energy.

Lighting occupancy sensors: Occupancy-based lighting controls in meeting rooms (106, 106A, 108) and the office (102) will reduce lighting energy during unoccupied periods. Lighting represents 35% of the building's electricity consumption and 12% of total site energy [see Table 9], making it the single most impactful electricity-saving measure.

11.5 Operational Costs

Annual operating costs are calculated using UVic's average utility rates: electricity at \$0.07/kWh and natural gas at \$13.9/GJ [6]. The eQUEST simulation results from Table 9 are used as the basis. Conversion: 1 THERM = 0.10551 GJ.

Annual electricity cost:

$$160,721 \text{ kWh/yr} \times \$0.07/\text{kWh} = \$11,250/\text{yr}$$

Annual natural gas cost:

$$10,524 \text{ THERM/yr} \times 0.10551 \text{ GJ/THERM} = 1,110 \text{ GJ/yr} \times \$13.9/\text{GJ} = \$15,434/\text{yr}$$

Total estimated annual utility cost:

$$\$11,250 + \$15,434 = \$26,684/\text{yr}$$

Heating (space heating + domestic hot water) accounts for \$15,434/yr or 57.9% of the total utility bill. Electricity for lighting accounts for \$3,937/yr (14.8%), ventilation fans for \$2,788/yr (10.5%), and cooling for \$1,376/yr (5.2%). The relatively low cooling cost reflects Victoria's mild summer climate.

If the ERV measure described in Section 11.3 is implemented, natural gas savings of approximately \$9,200/yr would bring total costs to approximately \$17,484/yr. If the PV array is also installed, electricity savings of approximately \$3,780/yr would further reduce costs to approximately \$13,704/yr - a reduction of 49% from the baseline.

12.0 Conclusion & Future Work

This report presents a summer peak cooling load analysis and HVAC design for The Grad House, a proposed renovation of the Halpern Centre for Graduate Students (Building #160) at the University of Victoria. The following principal results were obtained:

- The design peak cooling load is 39.5 kW (11.2 tons refrigeration) at 1% annual exceed outdoor conditions for Victoria, BC.
- Solar heat gain through west and southeast glazing (11,160 W) and lighting (8,164 W) are the two largest heat gain components, together accounting for over 60% of the sensible room load.
- The design supply airflow is 2,002 L/s with a 13°C supply air temperature. The minimum outdoor air requirement per ASHRAE 62.1-2019 is 554 L/s (27.7% OA fraction).
- The supply duct system was sized using the equal friction method at 0.8 Pa/m. All duct sections are documented in the Excel workbook (Appendix B, Figure B13) with drawing callout labels for Drawing A5.

- EH Price SPD supply diffusers are specified across 15 rooms, with throw checks confirming T0.25/L within the 0.5-1.5 acceptable range for all non-minimum-flow rooms.
- eQUEST simulation indicates annual site energy consumption of 1,601.0 MBTU/yr, with space heating (55.4%) being the dominant end use. Annual utility costs are estimated at \$26,684/yr based on UVic utility rates.
- Manual and eQUEST cooling energy estimates agree within 0.1%, providing mutual validation.

The following work is required to complete the HVAC design:

- Heating coil sizing: Perform a winter heat loss analysis and size the AHU heating coil for district energy hot water supply.
- Condensate sizing: Size the AHU condensate drain based on the latent cooling load.
- Room 110 ACH: Evaluate supplemental recirculating fan-coil units within the Restaurant/Lounge to achieve the minimum 8 ACH circulation rate required by ASHRAE for dining occupancies.
- Controls documentation: Develop a complete sequence of operations for the AHU based on the strategy outlined in Section 12.4.
- Energy model refinement: Update the eQUEST model with a 7-day operation schedule and actual equipment loads from the bar and restaurant specification once finalized.

13.0 References

- [1] ASHRAE, Handbook of Fundamentals (SI), Atlanta, GA: ASHRAE, 2013.
- [2] ASHRAE, Handbook of Fundamentals (SI), Atlanta, GA: ASHRAE, 2021.

- [3] ASHRAE, Handbook of Fundamentals (SI), Atlanta, GA: ASHRAE, 2001.
- [4] ASHRAE, ANSI/ASHRAE/IES Standard 90.1-2019: Energy Standard for Buildings Except Low-Rise Residential Buildings, Atlanta, GA: ASHRAE, 2019.
- [5] ASHRAE, ANSI/ASHRAE Standard 62.1-2019: Ventilation and Acceptable Indoor Air Quality, Atlanta, GA: ASHRAE, 2019.
- [6] University of Victoria, "MECH 442 Final Project RFP - 2026: RFP to Convert Existing Graduate Students Centre to a New Student Space," Victoria, BC: University of Victoria, 2026.
- [7] ASHRAE, Principles of Heating, Ventilating, and Air Conditioning, 7th ed., Atlanta, GA: ASHRAE, 2013.
- [8] EH Price, SPD Square Plaque Diffuser Product Catalogue, Edmonton, AB: EH Price Limited, 2023.
- [9] University of Victoria, "MECH 442 Diffuser Selection Handout," Victoria, BC: University of Victoria, January 2026.
- [10] ASHRAE, ANSI/ASHRAE Standard 55-2020: Thermal Environmental Conditions for Human Occupancy, Atlanta, GA: ASHRAE, 2020.
- [11] L. Carinelli, "MECH442_Heat_Gain_Calculations.xlsx - Summer Cooling Load Calculation Workbook," University of Victoria, Victoria, BC, March 2026.

Appendix A – AutoCAD Drawings

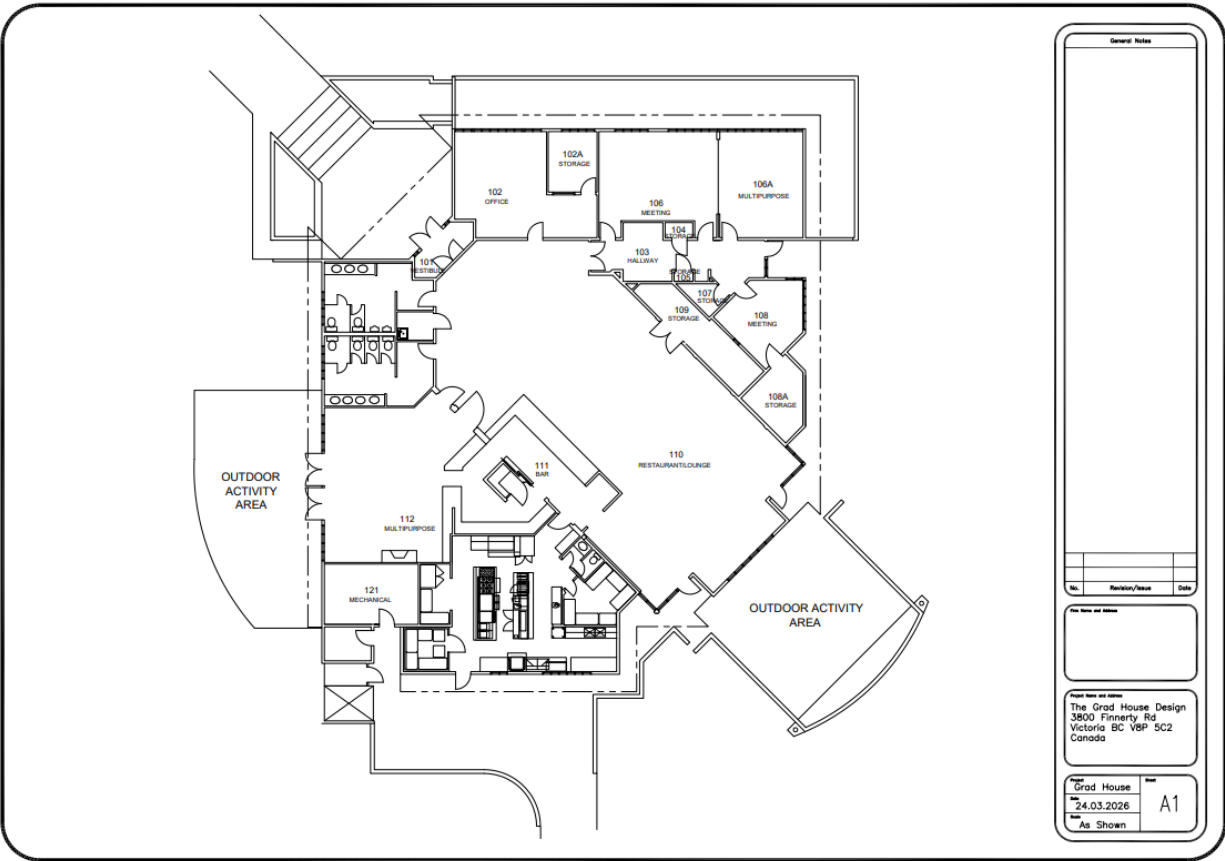


Figure A1: Drawing A1 - Existing Plan (Building #160)

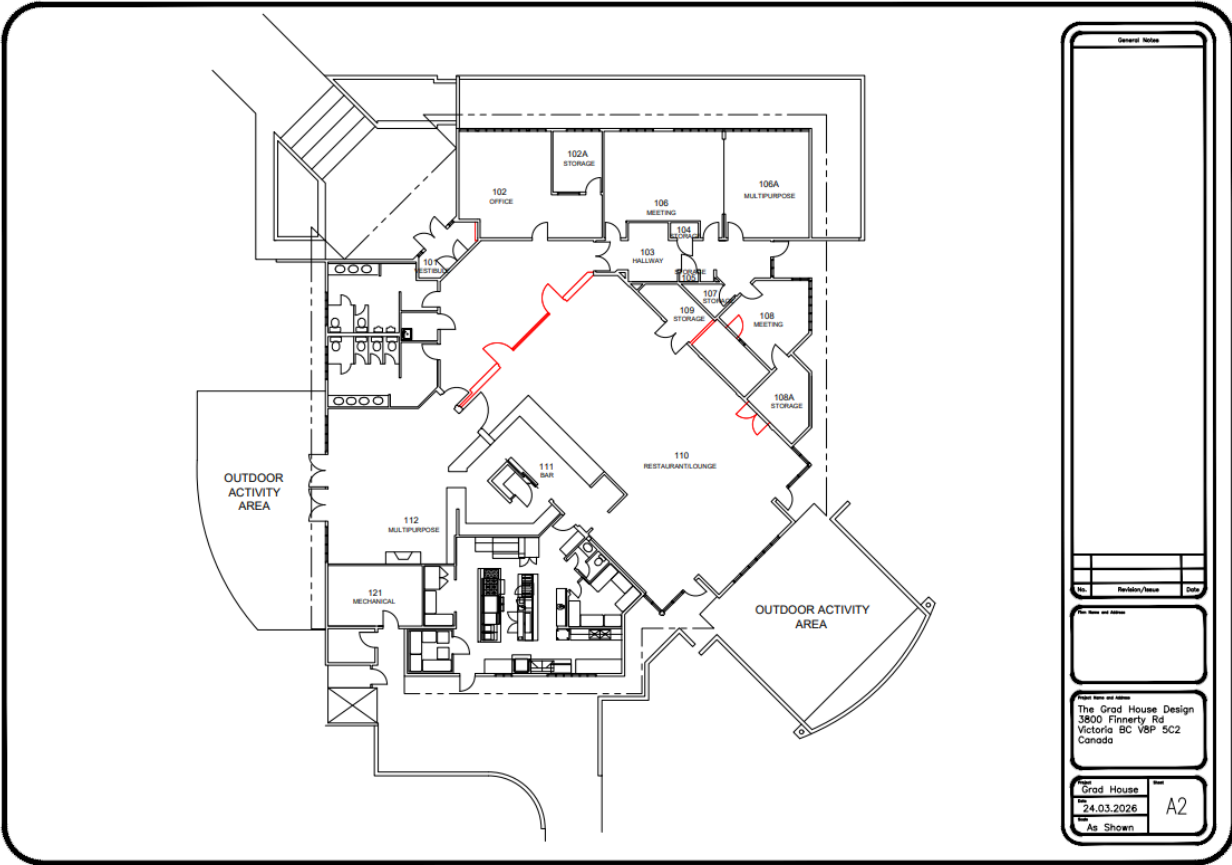


Figure A2: Drawing A2 - Demolition Plan

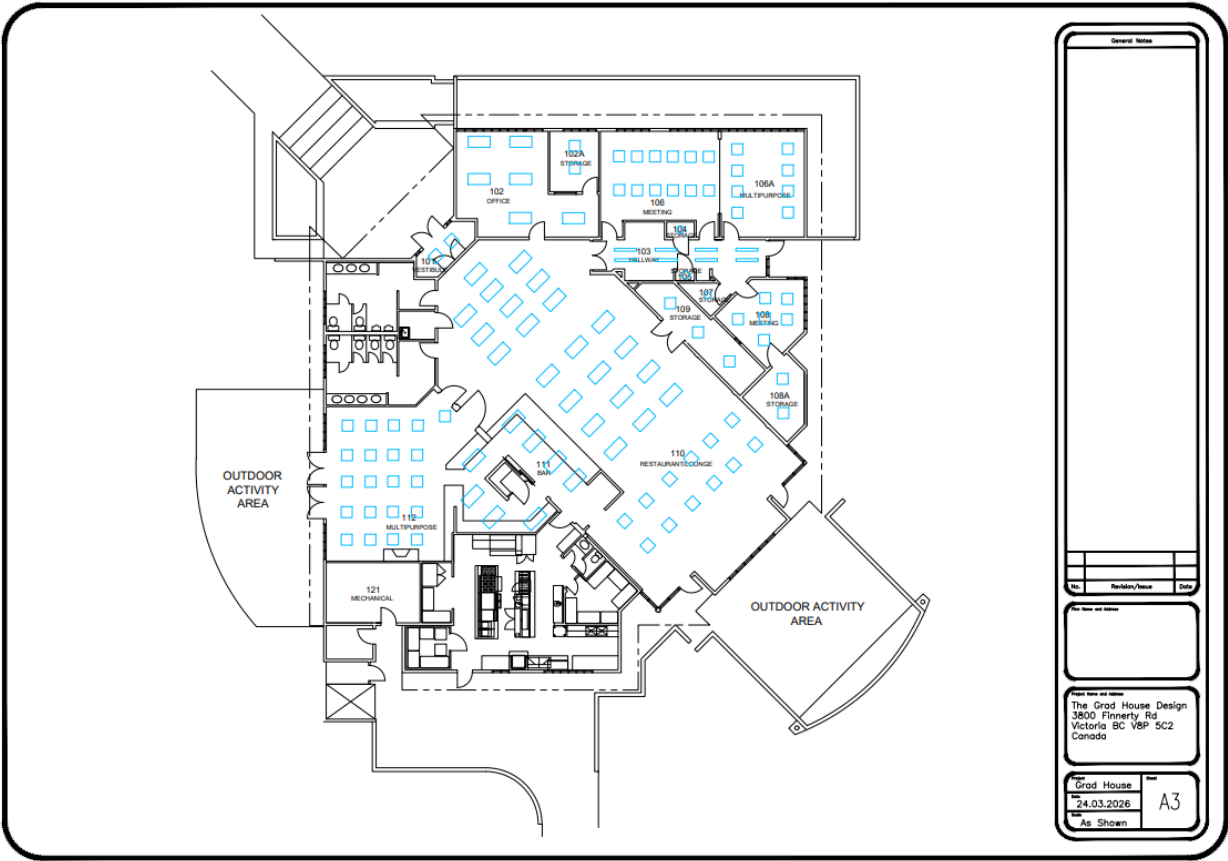


Figure A3: Drawing A3 - Lighting Plan

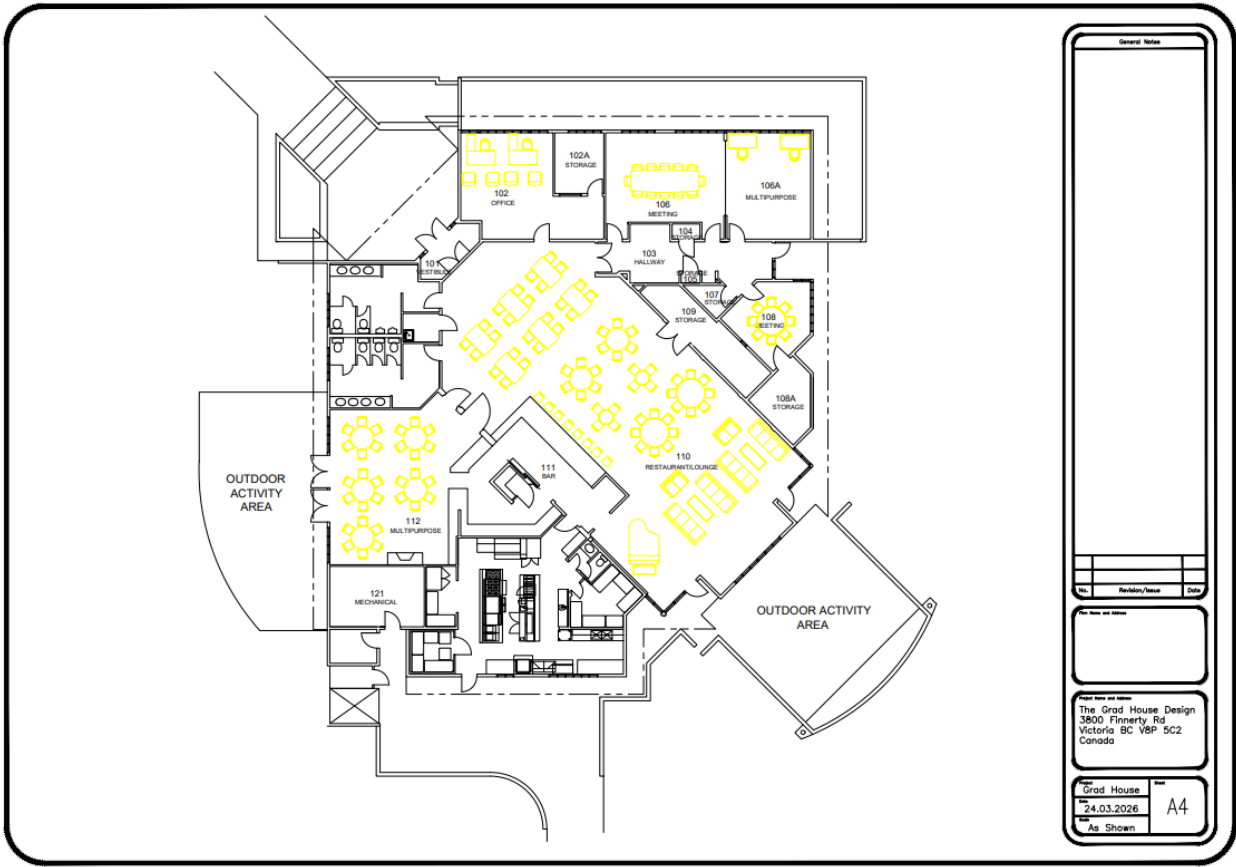


Figure A4: Drawing A4 - Interior Plan

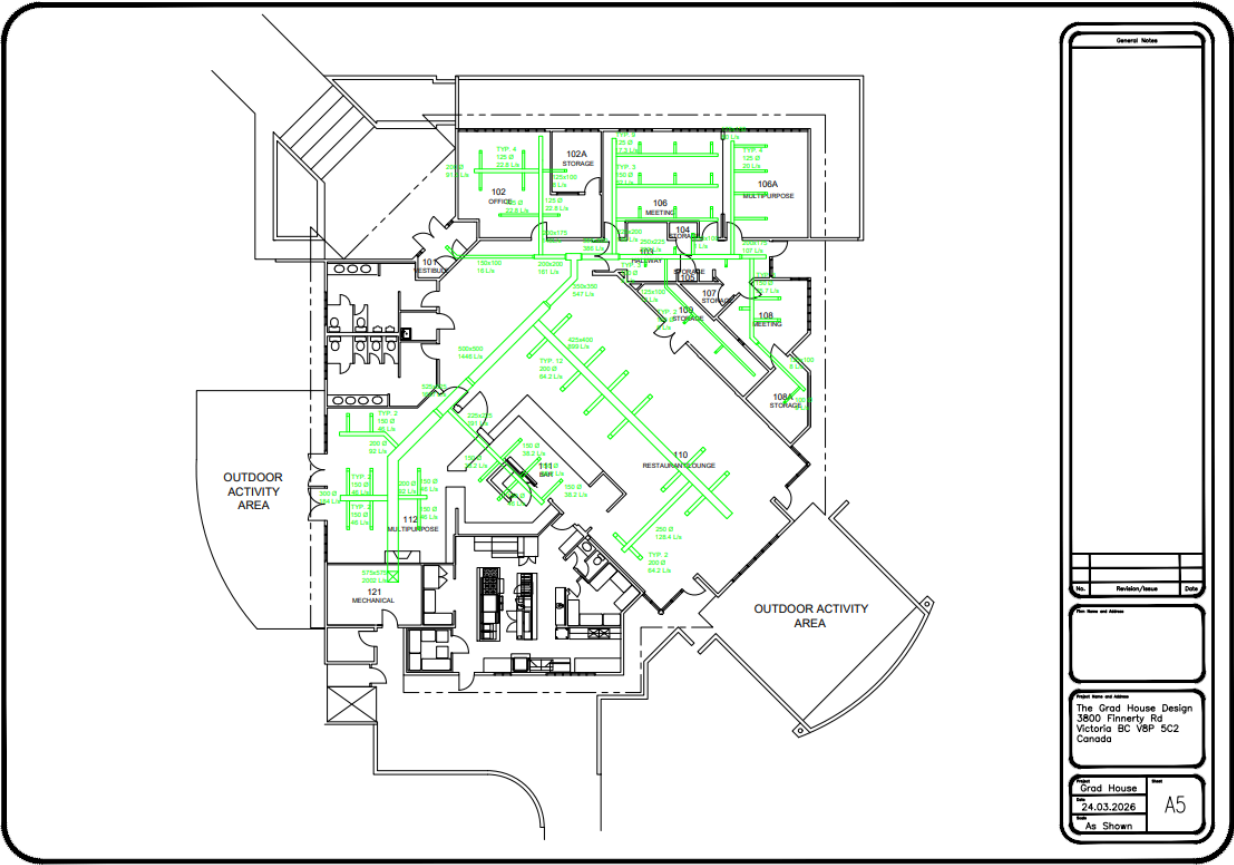


Figure A5: Drawing A5 - HVAC Plan (Supply Air Duct System)

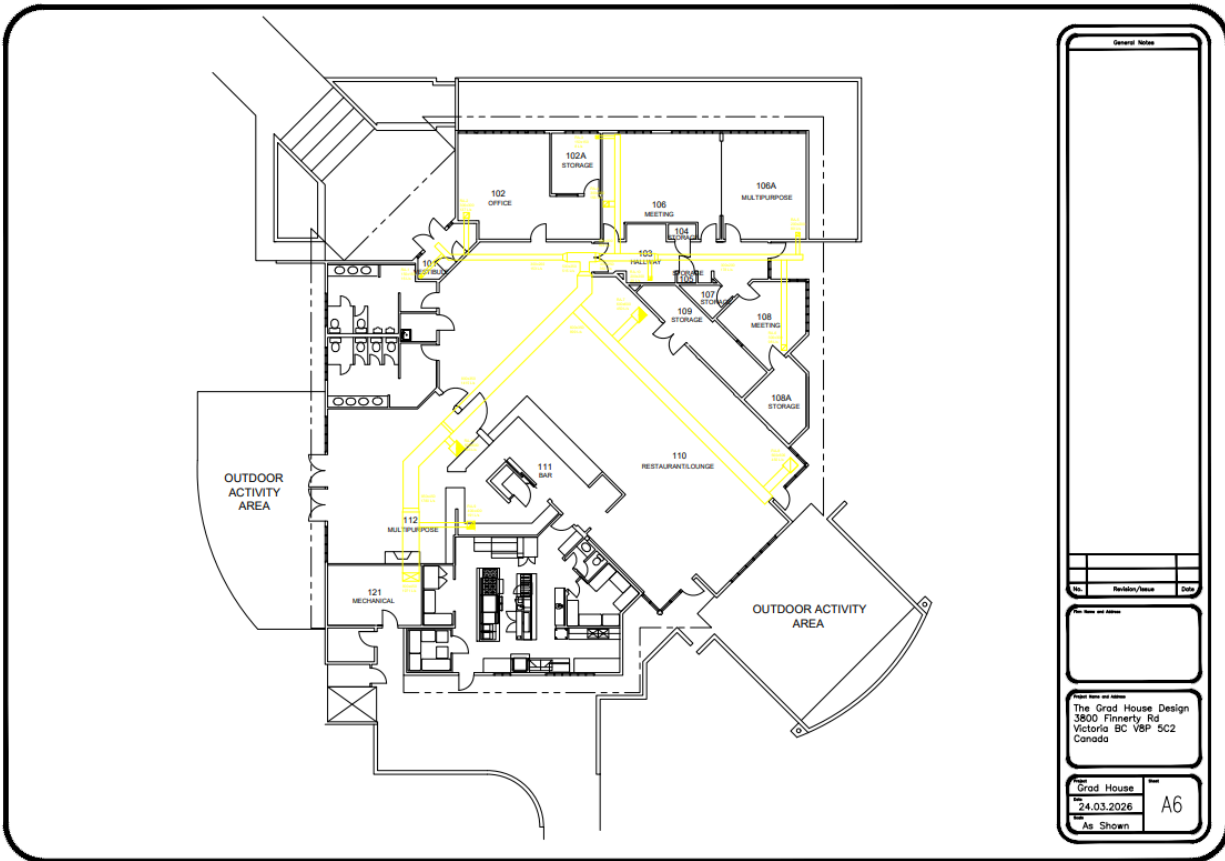


Figure A6: Drawing A6 - HVAC Plan (Return Air Duct System)

Appendix B – Calculations

Detailed cooling load calculations are contained in the Excel workbook [11]. The workbook contains the following sheets:

- Design Conditions and Areas: Room areas, ceiling heights, outdoor and indoor design conditions, and construction U-values. Sources: ASHRAE 2021 Fundamentals [2], ASHRAE 90.1-2019 [4].

- Internal Heat Gains: Room-by-room occupancy, lighting, and equipment heat gains.
Sources: ASHRAE 2013 Fundamentals Table 7-14 SI [1], ASHRAE 90.1-2010 Table 7-16 [4], ASHRAE 2013 Table 7-21A [1], ASHRAE 2001 Tables 8-10 [3].
- Envelope Heat Gains: Opaque wall and roof gains via sol-air temperature method, window conductive gains, and solar heat gains. Sources: ASHRAE 2013 Fundamentals Ch.18 [1], ASHRAE 90.1-2019 [4].
- Infiltration and Ventilation: ACH-based infiltration calculation and ASHRAE 62.1-2019 minimum outdoor air calculation by zone. Sources: ASHRAE 2013 Fundamentals Ch.16 [1], ASHRAE 62.1-2019 [5].
- Total Cooling Load Summary: Sum of all heat gain components with 10% safety factor, design load totals, supply airflow calculation, and OA fraction check.
- Supply Duct Sizing: Equal friction duct sizing for all 25 duct sections with drawing callout labels. Source: ASHRAE 2013 Fundamentals Ch.32 [1].
- Diffuser Selection: Step-by-step diffuser selection for all 15 rooms per the MECH 442 handout procedure [9] using the EH Price SPD catalogue [8].

MECH 442 - HVAC Load Calculations Halpern GSC New Student Space (Bldg #160), UVic Summer Cooling			
OUTDOOR SUMMER DESIGN CONDITIONS (Victoria BC - ASHRAE 2021 Fundamentals, Table 1, CZ 5C)			
Parameter	Value	Units	Reference / Notes
Outdoor Dry-Bulb Temp, 1% annual (T _{out})	26.6	°C	ASHRAE 2021 Fund., Victoria BC (48.65N 123.43W)
Outdoor Wet-Bulb Temp, 1% annual (T _{wb})	19.4	°C	ASHRAE 2021 Fund., Victoria Int'l A station
Outdoor Humidity Ratio (W _{out})	0.0104	kg _w /kg _{da}	Psychrometric chart @ 26.6°C DB / 19.4°C WB
Outdoor Enthalpy (h _{out})	52.8	kJ/kg	Psychrometric chart
Design Indoor Dry-Bulb (T _{in})	24	°C	ASHRAE 55-2020; Table 7-14 SI tabulated at 24°C
Design Indoor RH	50	%	ASHRAE 55-2020 comfort standard
Indoor Humidity Ratio (W _{in})	0.0093	kg _w /kg _{da}	Psychrometric @ 24°C / 50% RH
Indoor Enthalpy (h _{in})	47.7	kJ/kg	Psychrometric chart
dT conduction (T _{out} - T _{in})	2.6	°C	2.6
Supply Air Temperature (T _{supply})	13	°C	AHU cooling coil leaving air; dT _{supply} = 11°C below space
Sol-Air Temp, Roof (dark flat membrane)	55	°C	ASHRAE 2013 Ch.18 Eq.1: T _{sa} =T _{out} +alpha*I _t /h _o ; alpha=0.90, h _o =31
dT Sol-Air, Roof	31	°C	31
Sol-Air dT, North wall (diffuse only)	5.8	°C	I _t ~80 W/m², T _{sa} =26.6+0.9*80/22.7=29.8°C; ASHRAE 48°N July 21
Sol-Air dT, South wall	10.5	°C	I _t ~200 W/m², T _{sa} =34.5°C; low south gain in summer at 48°N
Sol-Air dT, East/West walls (peak)	30.4	°C	I _t ~700 W/m², T _{sa} =54.4°C; ASHRAE SHGF peak, July 21, 48°N
Sol-Air dT, SE wall (intermediate)	22.4	°C	I _t ~500 W/m², T _{sa} =46.4°C; interpolated SE orientation
SHGC glazing (double-pane LoE)	0.4	—	ASHRAE 90.1-2019 max CZ 5C; manufacturer ~0.38-0.42
Window U-value	2.56	W/m²K	ASHRAE 90.1-2019; double-pane LoE aluminum frame
Wall U-value (R-20 batt insulation)	0.28	W/m²K	ASHRAE 90.1-2019 CZ 5C prescriptive minimum
Roof U-value (R-30 rigid insulation)	0.19	W/m²K	ASHRAE 90.1-2019 CZ 5C prescriptive minimum
Outdoor surf. convection coeff. h _o	22.7	W/m²K	ASHRAE 2013 Ch.18; used in sol-air temp calculation
Surface absorptance alpha (exterior wall)	0.9	—	Dark paint; ASHRAE 2013 Table 7-1 footnote

Figure B1: Calculation Excel Sheet - Design Conditions and Area 1

ROOM AREAS & SPACE			
Room No.	Space Type	Floor Area (m²)	Ceiling Ht (m)
101	VESTIBULE	5.73	2.75
102	OFFICE	40	2.75
102A	STORAGE	8.74	2.75
103	HALLWAY	19.52	2.75
104	STORAGE	1.23	2.75
105	STORAGE	0.54	2.75
106	MEETING	33.19	2.75
106A	MULTIPURPOSE	26.05	2.75
107	STORAGE	2.79	2.75
108	MEETING	15.48	2.75
108A	STORAGE	9.48	2.75
109	STORAGE	14.29	2.75
110	RESTAURANT/LOUNGE	193.25	4.5
111	BAR	38.5	2.75
112	MULTIPURPOSE	58.18	2.75
TOTAL CONDITIONED FLOOR AREA		466.97	

Figure B2: Calculation Excel Sheet - Design Conditions and Area 2

Internal Heat Gains (Summer Cooling Load)									
2A. OCCUPANCY HEAT GAINS									
Room / Space Type	Activity (ASHRAE Table 7-14 SI)	Occ. Density (people/m²)	No. Persons	Sensible (W/person)	Latent (W/person)	Total Sensible (W)	Total Latent (W)	Total Heat (W)	ASHRAE Source
101 – VESTIBULE	Seated theater / light	10	1	70	30	70	30	100	Table 7-14 SI, ASHRAE 2013 Fund
102 – OFFICE	Seated very light work	7	6	70	45	420	270	690	Table 7-14 SI, ASHRAE 2013 Fund
102A – STORAGE	Unoccupied (storage)	0	0	0	0	0	0	0	Table 7-14 SI, ASHRAE 2013 Fund
103 – HALLWAY	Standing/walking	15	1	75	55	75	55	130	Table 7-14 SI, ASHRAE 2013 Fund
104 – STORAGE	Unoccupied (storage)	0	0	0	0	0	0	0	Table 7-14 SI, ASHRAE 2013 Fund
105 – STORAGE	Unoccupied (storage)	0	0	0	0	0	0	0	Table 7-14 SI, ASHRAE 2013 Fund
106 – MEETING	Seated very light work	3	11	70	45	770	495	1265	Table 7-14 SI, ASHRAE 2013 Fund
106A – MULTIPURPOSE	Seated very light work	6	4	70	45	280	180	460	Table 7-14 SI, ASHRAE 2013 Fund
107 – STORAGE	Unoccupied (storage)	0	0	0	0	0	0	0	Table 7-14 SI, ASHRAE 2013 Fund
108 – MEETING	Seated very light work	2	8	70	45	560	360	920	Table 7-14 SI, ASHRAE 2013 Fund
108A – STORAGE	Unoccupied (storage)	0	0	0	0	0	0	0	Table 7-14 SI, ASHRAE 2013 Fund
109 – STORAGE	Unoccupied (storage)	0	0	0	0	0	0	0	Table 7-14 SI, ASHRAE 2013 Fund
110 – RESTAURANT/LOUNGE	Sedentary work/restaurant	3	64	80	80	5120	5120	10240	Table 7-14 SI, ASHRAE 2013 Fund
111 – BAR	Standing light work	4	10	75	55	750	550	1300	Table 7-14 SI, ASHRAE 2013 Fund
112 – MULTIPURPOSE	Seated very light work	2	29	70	45	2030	1305	3335	Table 7-14 SI, ASHRAE 2013 Fund
TOTAL OCCUPANCY HEAT GAINS						10075	8365	18440	

Figure B3: Calculation Excel Sheet - Internal Heat Gains 1

2B. LIGHTING HEAT GAINS								
Room / Space Type	ASHRAE Table 7-16 Classification	Floor Area (m²)	LPD (W/m²)	LPD (W/m²)	Installed Power (W) (Area x LPD x DF)	Radiant Fraction	Sensible Heat (W)	Notes
101 – VESTIBULE	Lobby (common space)	5.73	0.9	9.68751	55.5094323	0.59	55.5094323	ASHRAE 90.1-2010/ Table 7-16: LPD=0.9 W/m²; EF=1.00 for LED
102 – OFFICE	Office - enclosed (common space)	40	1.11	11.947929	477.91716	0.59	477.91716	ASHRAE 90.1-2010/ Table 7-16: LPD=1.11 W/m²; EF=1.00 for LED
102A – STORAGE	Storage (common space)	8.74	0.63	6.781257	59.26818618	0.59	59.26818618	ASHRAE 90.1-2010/ Table 7-16: LPD=0.63 W/m²; EF=1.00 for LED
103 – HALLWAY	Corridor/transition (common space)	19.52	0.66	7.104174	138.6734765	0.59	138.6734765	ASHRAE 90.1-2010/ Table 7-16: LPD=0.66 W/m²; EF=1.00 for LED
104 – STORAGE	Storage (common space)	1.23	0.63	6.781257	8.34094611	0.59	8.34094611	ASHRAE 90.1-2010/ Table 7-16: LPD=0.63 W/m²; EF=1.00 for LED
105 – STORAGE	Storage (common space)	0.54	0.63	6.781257	3.66187878	0.59	3.66187878	ASHRAE 90.1-2010/ Table 7-16: LPD=0.63 W/m²; EF=1.00 for LED
106 – MEETING	Conference/meeting/multipurpose (cor)	33.19	1.23	13.239597	439.4222244	0.59	439.4222244	ASHRAE 90.1-2010/ Table 7-16: LPD=1.23 W/m²; EF=1.00 for LED
106A – MULTIPURPOSE	Conference/meeting/multipurpose (cor)	26.05	1.23	13.239597	344.8915019	0.59	344.8915019	ASHRAE 90.1-2010/ Table 7-16: LPD=1.23 W/m²; EF=1.00 for LED
107 – STORAGE	Storage (common space)	2.79	0.63	6.781257	18.91970703	0.59	18.91970703	ASHRAE 90.1-2010/ Table 7-16: LPD=0.63 W/m²; EF=1.00 for LED
108 – MEETING	Conference/meeting/multipurpose (cor)	15.48	1.23	13.239597	204.9489616	0.59	204.9489616	ASHRAE 90.1-2010/ Table 7-16: LPD=1.23 W/m²; EF=1.00 for LED
108A – STORAGE	Storage (common space)	9.48	0.63	6.781257	64.28631636	0.59	64.28631636	ASHRAE 90.1-2010/ Table 7-16: LPD=0.63 W/m²; EF=1.00 for LED
109 – STORAGE	Storage (common space)	14.29	0.63	6.781257	96.90416253	0.59	96.90416253	ASHRAE 90.1-2010/ Table 7-16: LPD=0.63 W/m²; EF=1.00 for LED
110 – RESTAURANT/LOUNGE	Lounge/recreation (common space)	193.25	0.73	7.857647	1518.490283	0.59	1518.490283	ASHRAE 90.1-2010/ Table 7-16: LPD=0.73 W/m²; EF=1.00 for LED
111 – BAR	For bar lounge/leisure dining (bldg-spe	38.5	1.31	14.100709	542.8772965	0.59	542.8772965	ASHRAE 90.1-2010/ Table 7-16: LPD=1.31 W/m²; EF=1.00 for LED
112 – MULTIPURPOSE	Conference/meeting/multipurpose (cor)	58.18	1.23	13.239597	770.2797535	0.59	770.2797535	ASHRAE 90.1-2010/ Table 7-16: LPD=1.23 W/m²; EF=1.00 for LED
TOTAL LIGHTING HEAT GAINS								4744.391286

Figure B4: Calculation Excel Sheet - Internal Heat Gains 2

2C. EQUIPMENT & APPLIANCE HEAT GAINS									
Equipment Item / Location	Qty	Rated Power (W/unit)	Usage Factor FU	Radiation Factor FR	Load Factor FL=FUxFR	Sensible Heat (W) [QtyxRatedxFLxFR]	Latent Heat (W)	Total (W)	ASHRAE Source
Coffee brewing urn (bar 111)	2	3010	0.08	0.17	0.0136	103.632	32.8	136.432	Table 7-23A SI
Reach-in refrigerator (bar 111)	1	1407	0.25	0.25	0.0625	87.9375	0	87.9375	Table 7-23A SI
Food warmer steam 2-well (lounge 110)	1	1495	0.69	0.08	0.0552	82.524	525.78	608.304	Table 7-23A SI
Hot-food case dry holding (lounge 110)	1	9115	0.11	0.27	0.0297	270.7155	19.36	290.0755	Table 7-23A SI
Drawer warmers 2-drawer (lounge 110)	2	1202	0.12	0	0	0	0	0	Table 7-23A SI
Laptop/desktop computers (office 102)	6	65	0.75	0.15	0.1125	43.875	0	43.875	ASHRAE 2001/2013 Table 8
Monitors medium 400-460mm (office 102)	6	70	0.75	0.38	0.285	119.7	0	119.7	ASHRAE 2001/2013 Table 9
Laser printer small desktop (office 102)	1	130	0.33	0.15	0.0495	6.435	0	6.435	ASHRAE 2001/2013 Table 10
AV projector (106, 106A, 108, 112)	4	300	0.5	0.25	0.125	150	0	150	Estimated 300W/rated
Cold food/beverage vending machine (106, 106A, 108, 112)	1	960	1	0	0	0	0	0	ASHRAE 2001 Table 10
Miscellaneous small appliances / table	1	400	0.4	0.2	0.08	32	0	32	
TOTAL EQUIPMENT HEAT GAINS						896.819	577.94	1474.759	

Figure B5: Calculation Excel Sheet - Internal Heat Gains 3

Envelope Heat Gains (Summer Cooling Load)						
3A. OPAQUE WALL HEAT GAIN						
North: $I_t \sim 80$, $dT=5.8^\circ\text{C}$ East/West: $I_t \sim 700$, $dT=30.4^\circ\text{C}$ SE: $I_t \sim 500$, $dT=22.4^\circ\text{C}$ South: $I_t \sim 200$, $dT=10.5^\circ\text{C}$						
Sol-air dT derivation: $T_{sa} = T_{out} + \alpha \cdot I_t / h_o$ (vertical, $R/h_o=0$). $T_{out}=26.6^\circ\text{C}$, $T_{in}=24.0^\circ\text{C}$						
Orientation / Room	Wall Gross Area (m ²)	Window Area (m ²)	Net Opaque Wall Area (m ²)	U-value (W/m ² K)	dT Sol-Air (°C)	Heat Gain Q (W)
North - Rm 102 (4.99m × 2.75m)	13.7225	7.0832	6.6393	0.28	5.8	10.7822232
North - Rm 102A (2.66m × 2.75m)	7.315	2.5992	4.7158	0.28	5.8	7.6584592
North - Rm 106 (6.4m × 2.75m)	17.6	6.5816	11.0184	0.28	5.8	17.8938816
North - Rm 106A (4.52m × 2.75m)	12.43	6.8704	5.5596	0.28	5.8	9.0287904
East - Rm 106A (5.76m × 2.75m)	15.84	0	15.84	0.28	30.4	134.83008
East - Rm 103 (2.02m × 2.75m)	5.555	1.2616	4.2934	0.28	30.4	36.5454208
East - Rm 108 (2.83m × 2.75m)	7.7825	4.3016	3.4809	0.28	30.4	29.6294208
East - Rm 108A (2.88m × 2.75m)	7.92	0	7.92	0.28	30.4	67.41504
West - Rm 112 (8.41m × 2.75m)	23.1275	5.5936	17.5339	0.28	30.4	149.2485568
SE - Rm 110 (11.16m × 4.5m)	50.22	9.4544	40.7656	0.28	22.4	255.6818432
TOTAL WALLS HEAT GAIN						718.713716

Figure B6: Calculation Excel Sheet - Envelop Heat Gains 1

3B. ROOF HEAT GAIN					
Roof Section	Roof Area (m ²)	U-value (W/m ² K)	dT Sol-Air (°C)	Q _{roof} (W)	Notes
Main conditioned roof	466.97	0.19	31	2750.4533	$Q = U \times A_{roof} \times dT_{sol-air, roof} \mid T_{sa}$
TOTAL ROOF HEAT GAIN				2750.4533	

Figure B7: Calculation Excel Sheet - Envelop Heat Gains 2

3C. WINDOW CONDUCTIVE HEAT GAIN						
All windows: 1.52 m height assumption.						
Orientation / Room	Window Width (m)	Window Height (m)	Window Area (m ²)	U-value (W/m ² K)	dT (°C)	Q _{cond} (W)
North - Rm 102 (4.66m × 1.52m)	4.66	1.52	7.0832	2.56	2.6	47.1457792
North - Rm 102A (1.71m × 1.52m)	1.71	1.52	2.5992	2.56	2.6	17.3002752
North - Rm 106 (4.33m × 1.52m)	4.33	1.52	6.5816	2.56	2.6	43.8071296
North - Rm 106A (4.52m × 1.52m)	4.52	1.52	6.8704	2.56	2.6	45.7293824
East - Rm 103 (0.83m × 1.52m)	0.83	1.52	1.2616	2.56	2.6	8.3972096
East - Rm 108 (2.83m × 1.52m)	2.83	1.52	4.3016	2.56	2.6	28.6314496
West - Rm 112 (3.68m × 1.52m)	3.68	1.52	5.5936	2.56	2.6	37.2310016
SE - Rm 110 (6.22m × 1.52m)	6.22	1.52	9.4544	2.56	2.6	62.9284864
TOTAL WINDOW HEAT GAIN						291.1707136

Figure B8: Calculation Excel Sheet - Envelop Heat Gains 3

3D. SOLAR HEAT GAIN THROUGH WINDOWS					
Orientation / Room	Window Area (m²)	SHGC (ASHRAE 90.1-2019)	SHGF (W/m²) 48°N July 21	Q_solar (W)	Notes
North - Rm 102	7.0832	0.4	100	283.328	
North - Rm 102A	2.5992	0.4	100	103.968	
North - Rm 106	6.5816	0.4	100	263.264	
North - Rm 106A	6.8704	0.4	100	274.816	
East - Rm 103	1.2616	0.4	480	242.2272	
East - Rm 108	4.3016	0.4	480	825.9072	
West - Rm 112	5.5936	0.4	480	1073.9712	
SE - Rm 110	9.4544	0.4	350	1323.616	
TOTAL SOLAR HEAT GAIN				4391.0976	

Figure B9: Calculation Excel Sheet - Envelop Heat Gains 4

Infiltration & Ventilation (Summer Cooling Load)			
4A. INFILTRATION HEAT GAIN			
Parameter	Value	Units	Notes / Reference
Total conditioned floor area	466.97	m²	
Weighted avg ceiling height	2.92	m	
Total conditioned volume (Vol)	1363.5524	m³	
Air change rate (ACH)	0.25	h⁻¹	
Infiltration flow rate Q_inf	0.094691139	m³/s	
Outdoor DB (T_out)	26.6	°C	
Indoor DB (T_in)	24	°C	
dT (T_out - T_in)	2.6	°C	
W_out (outdoor humidity ratio)	0.0104	kg/kg	Psychrometric @ 26.6°C / 19.4°C WB
W_in (indoor humidity ratio)	0.0093	kg/kg	Psychrometric @ 24°C / 50% RH
dW (W_out - W_in)	0.0011	kg/kg	
Air density rho	1.2	kg/m³	
Specific heat c_p	1006	J/kg·K	
Sensible infiltration Q_s	297.2089715	W	$Q_s = Q_{inf} \times \rho \times c_p \times dT$
Latent infiltration Q_l	312.6057506	W	$Q_l = Q_{inf} \times \rho \times h_{fg} \times dW$ ($h_{fg}=2501$ kJ/kg)
Total infiltration heat gain	609.8147221	W	Sensible + Latent

Figure B10: Calculation Excel Sheet - Infiltration and Ventilation 1

4B. MINIMUM VENTILATION REQUIREMENTS					
Zone / Rooms	Occupants (design)	R_p (L/s/person)	R_a (L/s/m²)	Zone Area (m²)	Total OA (L/s)
Office 102	6	5	0.06	40	32.4
Hallway 103 + Vestibule 101	2	0	0.06	25.25	1.515
Meeting 106	11	5	0.06	33.19	56.9914
Multipurpose 106A	4	5	0.06	26.05	21.563
Meeting 108	8	5	0.06	15.48	40.9288
Restaurant/Lounge 110	64	3.8	0.06	193.25	254.795
Bar 111	10	3.8	0.06	38.5	40.31
Multipurpose 112	29	5	0.06	58.18	148.4908
TOTAL MINIMUM OUTDOOR AIR					596.994

Figure B11: Calculation Excel Sheet - Infiltration and Ventilation 2

TOTAL COOLING LOAD SUMMARY				
Halpern GSC Building #160, UVic Summer Peak Cooling Load				
Heat Gain Component	Sensible (W)	Latent (W)	Total (W)	
Occupancy — Sensible	10075	0	10075	
Occupancy — Latent	0	8365	8365	
Lighting	4744.391286	0	4744.391286	
Equipment — Sensible	896.819	0	896.819	
Equipment — Latent	0	577.94	577.94	
Envelope — Walls	718.713716	0	718.713716	
Envelope — Roof	2750.4533	0	2750.4533	
Envelope — Windows Cond.	291.1707136	0	291.1707136	
Envelope — Windows Solar	4391.0976	0	4391.0976	
Infiltration — Sensible	297.2089715	0	297.2089715	
Infiltration — Latent	0	312.6057506	312.6057506	
SUBTOTAL ROOM LOAD	24164.85459	9255.545751	33420.40034	
Design Parameter	Value	Units	Formula / Notes	
Safety / diversity factor (SF)	1.1			
DESIGN TOTAL SENSIBLE LOAD	26581.34005	W'	$Q_{s,design} = Q_{s,room} \times SF$	
DESIGN TOTAL LATENT LOAD	10181.10033	W'	$Q_{l,design} = Q_{l,room} \times SF$	
DESIGN TOTAL COOLING LOAD	36762.44037	W'	$Q_{total} = Q_s + Q_l$	
Sensible Heat Ratio (SHR)	0.723057005		$SHR = Q_s / Q_{total}$; target 0.75-0.90	
SUPPLY AIR SIZING				
Supply air temperature T _s	13	°C	AHLU cooling coil leaving air	
dT supply (T _{room} - T _{supply})	11	°C		
Supply airflow Q _{supply}	2.001727517	m³/s	$Q = Q_s / (\rho \times c_p \times \Delta T)$	
Supply airflow Q _{supply}	2001.727517	L/s		
Minimum OA (ASHRAE 62.1-2019)	596.994	L/s	Must be delivered by AHLU	

Figure B12: Calculation Excel Sheet - Total Cooling Load Summary

SECTION B: DUCT SIZING SCHEDULE (SUPPLY AIR) $Q = \text{flow (m}^3/\text{s)} D = [(f \cdot p \cdot Q^3) / (2 \cdot (\pi/4)^2 \cdot \Delta P/L)]^{0.2} D_{eq} = 1.30 \cdot (W \cdot H)^{0.625} / (W+H)^{0.25}$							
Section ID	Routing / Description	Flow (L/s)	Rect W (mm)	Rect H (mm)	Round Ø (mm)	Velocity (m/s)	Friction s
MAIN TRUNK							
SA	AHU → entering Rm 112 (MAIN TRUNK)	2001.7	575	575	625	6.05	0.7
SB	After Rm 112 takeoff → toward Rm 111	1637	525	525	575	5.94	0.737
SC	After Rm 111 takeoff → toward Rm 110	1446.2	500	500	550	5.78	0.735
SD	After Rm 110 takeoff → T-junction	547.3	350	350	375	4.47	0.626
MAIN TRUNK TAKEOFFS							
T-112	Takeoff → Rm 112 Multipurpose (58.18 m')	364.8	300	300	325	4.05	0.601
T-111	Takeoff → Rm 111 Bar (38.50 m')	190.8	225	225	250	3.77	0.693
T-110	Takeoff → Rm 110 Restaurant/Lounge (193 m')	898.8	425	400	450	5.29	0.744
LEFT BRANCH (T-jct → 102 → 102A → 101)							
SR1	T-jct RIGHT → toward Rm 102 (right branch)	161	200	200	225	4.02	0.889
T-102	Takeoff → Rm 102 Office (40.0 m')	137.3	200	175	225	3.93	0.836
SR2	After Rm 102 takeoff → toward 102A + 101	23.7	150	125	150	1.26	0.117
T-102A	Takeoff → Rm 102A Storage (8.74 m')	7.6	125	100	100	0.61	0.04
SR3/T-101	Final run → Rm 101 Vestibule (5.73 m')	16.1	150	100	125	1.07	0.1
RIGHT BRANCH (T-jct → 103 → 106 → 109/104/105 → 106A / 108+108A)							
SL1	T-jct LEFT → toward Rm 103 (left branch)	386.3	300	300	325	4.29	0.674
T-103	Takeoff → Rm 103 Hallway (19.52 m')	27.5	150	100	150	1.83	0.285
SL2	After Rm 103 takeoff → toward Rm 106	358.8	300	275	325	4.34	0.714
T-106	Takeoff → Rm 106 Meeting (33.19 m')	155.5	225	200	225	3.44	0.609
SL3	After Rm 106 takeoff → toward 103/104/105	203.3	250	225	250	3.61	0.629
T-109	Takeoff → Rm 109 Storage (14.29 m')	12.5	125	100	125	1	0.112
SL4	After Rm 109 takeoff → toward 104/105	190.8	250	225	250	3.39	0.554
T-104	Takeoff → Rm 104 Storage (1.23 m') MIN DUCT	1.1	150	100	100	0.07	0.001
SL5	After Rm 104 takeoff → toward Rm 105	189.7	250	225	250	3.37	0.549
T-105	Takeoff → Rm 105 Storage (0.54 m') MIN DUCT	0.5	150	100	100	0.03	0
SL6	After Rm 105 → final T-jct (106A / 108+108A)	189.2	250	225	250	3.36	0.546
T-106A	Final run → Rm 106A Multipurpose (26.05 m')	80.3	175	150	175	3.06	0.634
SL7	Branch → toward Rm 108 + Rm 108A	106.6	200	175	200	3.04	0.559
T-108	Takeoff → Rm 108 Meeting (15.48 m')	98.3	200	175	200	2.8	0.475
T-108A	Final run → Rm 108A Storage (3.48 m')	8.3	125	100	100	0.66	0.049

Figure B13: Calculation Excel Sheet - Supply Duct Sizing

Rm	Space Type	Area (m ²)	Cell Ht (m)	Supply (L/s) From A5	ACH Table 4-4	ACH Q (L/s)	Gov. erns	Qty Diff (A5)	Q per Diff (L/s)	Char Len L (m)	SPD Frame mm	Listed Size mm	TP (Pa)	T _{hi} (m)	T/L Ratio	Branch Ø mm	Drawing Tag / Notes (from Drawing A5)
101	Vestibule	5.73	2.75	16	7	30.6	ACH	1	16	1.21	300x300	150	32	1.7	1.40	125	1x 1250 16 L/s A5: 150x100 16 L/s trunk → 1 diff EH Price SPD 300x300 / 150mm listed
102	Office	40	2.75	145	8	244.4	ACH	4	36.3	1.59	400x400	200	23	2.2	1.38	125	4x 1250 36.3 L/s each A5: 200x175 145 L/s trunk → > 4 diff EH Price SPD 400x400 / 200mm listed
102A	Storage	8.74	2.75	8	5	33.4	ACH	1	8	1.49	300x300	150	10	1.1	0.74	100	1x 1000 8 L/s A5: 125x100 8 L/s trunk SPD 300x300 / 150mm min flow room
103	Hallway	19.52	2.75	27	7	104.4	ACH	1	27	2.22	300x300	150	52	2.1	0.95	150	1x 1500 27 L/s Mass balance: SL1 386 - T106 156 - SL3 203 + 27 L/s SPD 300x300 / 150mm 1x 1000 1 L/s A5: min duct 150x100 1 L/s MIN FLOW - balancing damper near-closed
104	Storage	1.23	2.75	1	—	—	MIN	1	1	—	300x300	150	10	—	—	100	1x 1000 1 L/s A5: min duct MIN FLOW - balancing damper near-closed
105	Storage	0.54	2.75	1	—	—	MIN	1	1	—	300x300	150	10	—	—	100	1x 1000 1 L/s A5: min duct MIN FLOW - balancing damper near-closed
106	Meeting	33.19	2.75	156	8	202.8	ACH	3	52	1.67	500x500	250	16	2.3	1.38	150	3x 1500 52 L/s each A5: 150 O 52 L/s TYP 3 confirmed EH Price SPD 500x500 / 250mm listed
106A	Multipurpose	26.05	2.75	80	8	159.2	ACH	1	80	2.56	500x500	250	52	3.8	1.48	175	1x 1750 80 L/s A5: 175x150 80 L/s direct duct to 1 diff EH Price SPD 500x500 / 250mm
107	Storage (XFER)	2.79	2.75	0	—	—	—	—	—	—	—	—	—	—	—	—	TRANSFER GRILLE 200x100mm face - partition wall between Rm 107 and Rm 103 no supply diff
108	Meeting	15.48	2.75	92	8	94.6	ACH	2	46	1.40	500x500	250	10	1.8	1.29	150	2x 1500 46 L/s each A5: 200O 92 L/s feeder → 2x 1500 46 L/s TYP 2 SPD 500x500 / 250mm
108A	Storage	9.48	2.75	8	5	36.2	ACH	1	8	1.55	300x300	150	10	1.1	0.71	100	1x 1000 8 L/s A5: T-108A 125x100 8 L/s trunk SPD 300x300 / 150mm listed
109	Storage	14.29	2.75	12	5	54.6	ACH	1	12	1.90	300x300	150	16	1.3	0.68	125	1x 1250 12 L/s A5: 125x100 12 L/s trunk SPD 300x300 / 150mm listed
110	Rest./Lounge	193.3	4.5	899	12	3019.5	ACH	14	64.2	3.62	500x500	250	32	3.1	0.86	200	14x 2000 64.2 L/s A5: TYP 12 + TYP 2 2000 64.2 L/s (2500 128.4 intermediate) SPD 500x500 / 5x 1500 38.2 L/s each A5: 1500 38.2 L/s x3 confirmed EH Price SPD 400x400 / 200mm listed
111	Bar	38.5	2.75	191	12	367.6	ACH	5	38.2	1.40	400x400	200	23	2.2	1.57	150	4x 2000 91.3 L/s each A5: 2000 91.3 L/s TYP 4 confirmed EH Price SPD 500x500 / 250mm listed
112	Multipurpose	58.18	2.75	365	8	355.5	LOAD	4	91.3	1.92	500x500	250	64	3.8	1.98	200	4x 2000 91.3 L/s each A5: 2000 91.3 L/s TYP 4 confirmed EH Price SPD 500x500 / 250mm listed
TOTAL SUPPLY DIFFUSERS: 40 Room 107 = transfer grille (not counted) All flows and branch sizes from Drawing A5																	

Figure B14: Calculation Excel Sheet - Diffuser Selection

Velocity limits: Main trunk <=7.5 m/s Branch <=6.0 m/s Final run <=5.0 m/s Fittings per ASHRAE Fund. Ch.21 Vel. pressure method																
Duct Section	Flow (L/s)	Flow (m3/s)	D _{eq} mm	Ht mm	Wd mm	Act.Wd mm	Area m2	Vel m/s	VP Pa	Duct Label	L duct ft	L duct m	L fit m	L tot m	dP/L Pa/m	Sect dP Pa
--	L/s	m3/s	mm	mm	mm	mm	m2	m/s	Pa	--	ft	m	m	m	Pa/m	Pa
1 (SA)	2002	2.002	629	575	575	575	0.3306	6.06	22	SA	19.7	6	7.21	13.21	0.7	9.25
2 (SB)	1637	1.637	574	525	525	525	0.2756	5.94	21.16	SB	16.4	5	8.06	13.06	0.738	9.63
3 (SC)	1446	1.446	547	500	500	500	0.25	5.78	20.07	SC	23	7	9.58	16.58	0.734	12.18
4 (SD)	547	0.547	383	350	350	350	0.1225	4.47	11.96	SD	19.7	6	5.37	11.37	0.625	7.11
5 (SR1)	161	0.161	219	200	200	200	0.04	4.02	9.72	SR1	26.2	8	12.77	20.77	0.889	18.47
6 (T-102)	145	0.145	204	175	200	200	0.035	4.14	10.3	T-102	13.1	4	4.82	8.82	1.008	8.89
7 (T-101)	16	0.016	133	100	150	150	0.015	1.07	0.68	T-101	19.7	6	1.87	7.87	0.102	0.81
8 (SL1)	386	0.386	328	300	300	300	0.09	4.29	11.04	SL1	16.4	5	11.85	16.85	0.673	11.34
9 (SL3)	203	0.203	259	225	250	250	0.0563	3.61	7.81	SL3	16.4	5	5.9	10.9	0.603	6.57
10 (T-106A)	80	0.08	177	150	175	175	0.0262	3.05	5.57	T-106A	16.4	5	7.84	12.84	0.63	8.09
11 (SL7)	107	0.107	204	175	200	200	0.035	3.06	5.61	SL7	13.1	4	7.54	11.54	0.549	6.33
12 (T-108)	99	0.099	204	175	200	200	0.035	2.83	4.8	T-108	9.8	3	7.83	10.83	0.47	5.09
13 (T-108A)	8	0.008	122	100	125	125	0.0125	0.64	0.25	T-108A	13.1	4	25	29	0.04	1.17

Figure B15: Calculation Excel Sheet - Fan Sizing 1

			Fitting Loss Calculations - Velocity Pressure Method (ASHRAE Fund. Ch.21)						
Path 1-7	Path 1-10	Path 1-13							
Pa	Pa	Pa	Section	Fitting Types / Notes	Loss Coeff		Leq duct m	Leq diff m	Total Leq m
9.25	9.25	9.25	SA	1x elbow AHU outlet; 1x tee-through Rm112 n=0.05	n=0.05+el		5.64	1.57	7.21
9.63	9.63	9.63	SB	1x elbow routing; 1x tee-through Rm111 n=0.10	n=0.10+el		5.18	2.88	8.06
12.18	12.18	12.18	SC	Reducer 525->500 L/D=5; tee-through Rm110 n=0.25	n=0.25+red		2.74	6.84	9.58
7.11	7.11	7.11	SD	Reducer 500->350; 1x elbow to T-junction	n=0.05+red		1.92	3.45	5.37
18.47			SR1	Tee-BRANCH right at T-jct n=0.80; 1x elbow	n=0.80		1.97	10.8	12.77
8.89			T-102	Tee-branch SR1->Rm102 n=0.50; diff SP 400x400	n=0.50		0	4.82	4.82
0.81			T-101	Reducer + elbow; final low-vel run; diff 300x300	n=0.05		1.87	0	1.87
	11.34	11.34	SL1	Tee-BRANCH left at T-jct n=0.50; 2x elbows	n=0.50		2.95	8.9	11.85
	6.57	6.57	SL3	Tee-through Rm103 n=0.05; Rm106 n=0.20; reducer	n=0.20		2.22	3.68	5.9
	8.09		T-106A	Tee-branch SL3->106A n=0.50; 1x elbow; diff SP	n=0.50		1.6	6.24	7.84
		6.33	SL7	Tee-branch SL3->SL7 n=0.50; minor tees 109/104	n=0.50		0.41	7.13	7.54
		5.09	T-108	Tee-branch SL7->Rm108 n=0.50; 2x elbows; diff SP	n=0.50		1.85	5.98	7.83
		1.17	T-108A	Branch T-108->108A n=0.50 + elbows; diff SP 400x400	n=0.50		18.5	6.5	25

Figure B16: Calculation Excel Sheet - Fan Sizing 2

				Diffuser Static Pressures (EH Price SPD Catalogue)					
	PATH PRESSURE SUMMARY			Paths 1-7 and 1-10 (SPD at design flow):					
	Path 1->7 (->Rm101)	Path 1->10 (->Rm106A)	Path 1->13 (->Rm108A)	Path 1-13 (400x400/200mm at 8 L/s):					
Section pressure drops (Pa):	66.34	64.17	68.67						
Velocity regain (Pa):	-1.68	-2.42	-4.07						
Diffuser total pressure (Pa):	10	10	23						
PATH TOTAL REQUIRED (Pa):	74.7	71.8	87.6						
	FAN SPECIFICATION: Flowrate = 2,002 L/s Ext. Static Pressure = 88 Pa Critical path: AHU to Rm 108A (Path 1->13)								

Figure B16: Calculation Excel Sheet - Fan Sizing 3

Appendix C – eQUEST Reports

MECH 442 Final Project - The Grad House						DOE-2.2-50a		3/28/2026		13:03:10		BDL RUN 1	
REPORT- BEPS Building Energy Performance						WEATHER FILE- Victoria Internation							
	LIGHTS	TASK LIGHTS	MISC EQUIP	SPACE HEATING	SPACE COOLING	HEAT REJECT	PUMPS & AUX	VENT FANS	REFRIG DISPLAY	HT PUMP SUPPLEM	DOMEST HOT WTR	E/T USAGE	TOTAL

EMI ELECTRICITY													
MBTU	191.9	0.0	152.8	0.0	67.1	0.0	0.8	135.9	0.0	0.0	0.0	0.0	548.5
FMI NATURAL-GAS													
MBTU	0.0	0.0	0.0	887.6	0.0	0.0	0.0	0.0	0.0	0.0	164.9	0.0	1052.4
=====													
MBTU	191.9	0.0	152.8	887.6	67.1	0.0	0.8	135.9	0.0	0.0	164.9	0.0	1601.0
TOTAL SITE ENERGY				1600.98 MBTU		111.0 KBTU/SQFT-YR GROSS-AREA		111.0 KBTU/SQFT-YR NET-AREA					
TOTAL SOURCE ENERGY				2698.06 MBTU		187.1 KBTU/SQFT-YR GROSS-AREA		187.1 KBTU/SQFT-YR NET-AREA					
PERCENT OF HOURS ANY SYSTEM BONE OUTSIDE OF THROTTLING RANGE =										0.02			
PERCENT OF HOURS ANY PLANT LOAD NOT SATISFIED										= 0.00			
HOURS ANY BONE ABOVE COOLING THROTTLING RANGE										= 0			
HOURS ANY BONE BELOW HEATING THROTTLING RANGE										= 1			
NOTE: ENERGY IS APPORTIONED HOURLY TO ALL END-USE CATEGORIES.													
MECH 442 Final Project - The Grad House						DOE-2.2-50a		3/28/2026		13:03:10		BDL RUN 1	
REPORT- BEPU Building Utility Performance						WEATHER FILE- Victoria Internation							
	LIGHTS	TASK LIGHTS	MISC EQUIP	SPACE HEATING	SPACE COOLING	HEAT REJECT	PUMPS & AUX	VENT FANS	REFRIG DISPLAY	HT PUMP SUPPLEM	DOMEST HOT WTR	E/T USAGE	TOTAL

EMI ELECTRICITY													
KWH	56237.	0.	44758.	0.	19650.	0.	245.	39832.	0.	0.	0.	0.	160721.
FMI NATURAL-GAS													
THERM	0.	0.	0.	8876.	0.	0.	0.	0.	0.	0.	1649.	0.	10524.
TOTAL ELECTRICITY			160721. KWH		11.147 KWH /SQFT-YR GROSS-AREA		11.147 KWH /SQFT-YR NET-AREA						
TOTAL NATURAL-GAS			10524. THERM		0.730 THERM /SQFT-YR GROSS-AREA		0.730 THERM /SQFT-YR NET-AREA						
PERCENT OF HOURS ANY SYSTEM BONE OUTSIDE OF THROTTLING RANGE =										0.02			
PERCENT OF HOURS ANY PLANT LOAD NOT SATISFIED										= 0.00			
HOURS ANY BONE ABOVE COOLING THROTTLING RANGE										= 0			
HOURS ANY BONE BELOW HEATING THROTTLING RANGE										= 1			
NOTE: ENERGY IS APPORTIONED HOURLY TO ALL END-USE CATEGORIES.													

Figure C1: BEPS and BEPU Results - eQuest